

LECTURE 3

Linear classification

Foundations of Artificial Intelligence

SEMTM0051 · 2026/27

Lecturer: Dr Roberto Colomboni



Today's lecture

Binary classification

Classification error and logistic loss

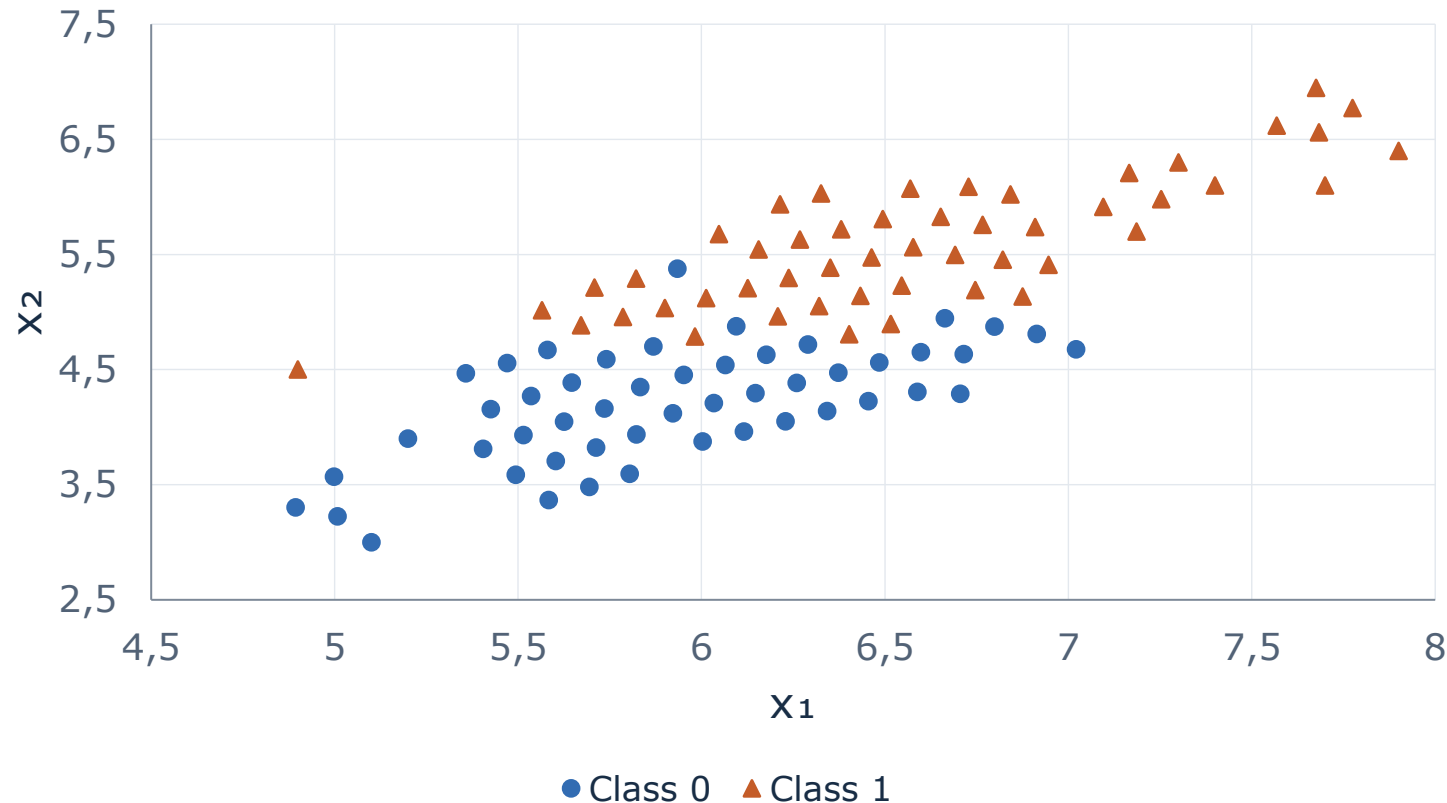
Gradient descent

More than two classes

Binary classification

A synthetic dataset adapted from Iris:

100 observations $(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_{100}, y_{100})$, with two input features and two classes.



$$\mathbf{x}_i = (x_{i1}, x_{i2}), \quad y_i \in \{0,1\}$$

x_1 : first input feature

x_2 : second input feature

Class 0: 50 observations

Class 1: 50 observations

**Learn a rule to predict
the class of an input.**

[Dataset and instructions in the folder: Week 3 - Examples](#)

Measuring classification error

A regression **predictor** $f: \mathbb{R}^d \rightarrow \mathbb{R}$
returns a **number**

Regression

$$\ell_{\text{sq}}(\hat{y}, y) = (\hat{y} - y)^2$$

Empirical squared loss

$$\sum_{i=1}^N \ell_{\text{sq}}(f(\mathbf{x}_i), y_i)$$

Larger numerical errors
receive larger penalties

A **classifier** $h: \mathbb{R}^d \rightarrow \{0,1\}$
returns a class **label**.

Classification

$$\ell_{0,1}(\hat{y}, y) = \begin{cases} 0, & \hat{y} = y \\ 1, & \hat{y} \neq y \end{cases}$$

Number of mistakes / Empirical 0-1 loss

$$\sum_{i=1}^N \ell_{0,1}(h(\mathbf{x}_i), y_i)$$

All mistakes are penalized
the same way

Predicting a probability first

A label gives a **decision**.

A **probability** also records how strongly the model favours that label.

Probabilistic predictor

$$p : \mathbb{R}^d \rightarrow [0,1]$$

Probability of class 1

$$p(\mathbf{x})$$

Probability of class 0

$$1 - p(\mathbf{x})$$

Example: if $p(\mathbf{x}) = 0.51$ we think that there is

- a 0.51 probability that the correct class of input $\mathbf{x}$ is 1
- a 0.49 probability that the correct class of input $\mathbf{x}$ is 0

If we must choose one label, which should we predict?

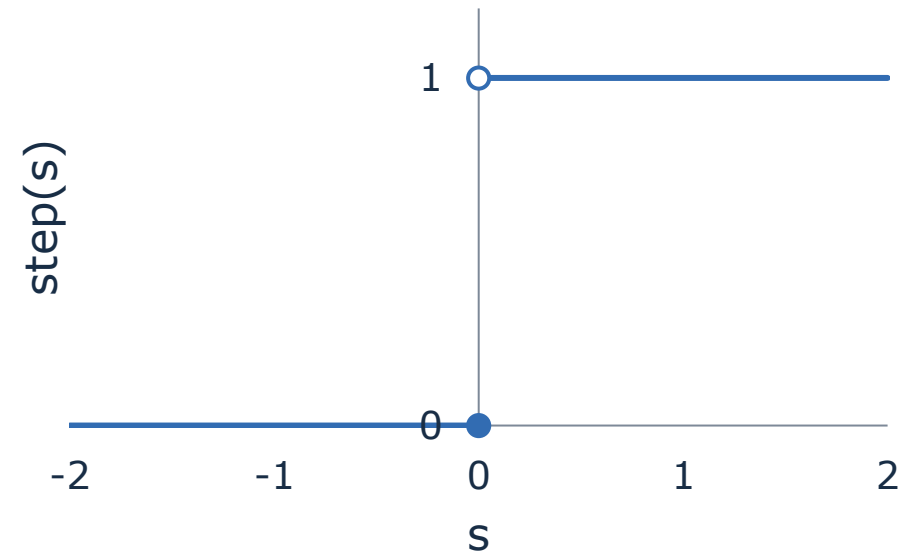
The plug-in classification rule

Choose the label with the larger probability. At a tie, we choose class 0.

$$h_p(\mathbf{x}) = \begin{cases} 0, & p(\mathbf{x}) \leq 1/2 \\ 1, & p(\mathbf{x}) > 1/2 \end{cases}$$

Heaviside step function

$$\text{step}(s) := \begin{cases} 0, & s \leq 0, \\ 1, & s > 0. \end{cases}$$



$$h_p(\mathbf{x}) = \text{step}\left(p(\mathbf{x}) - \frac{1}{2}\right)$$

How do we choose the probability function?

How do we build probability predictions?

Regression suggests using an intercept and one weight for each input feature.

$$f_{\boldsymbol{\theta}}(\mathbf{x}) = b + \mathbf{x}\mathbf{w} = b + \sum_{j=1}^d x_j w_j$$

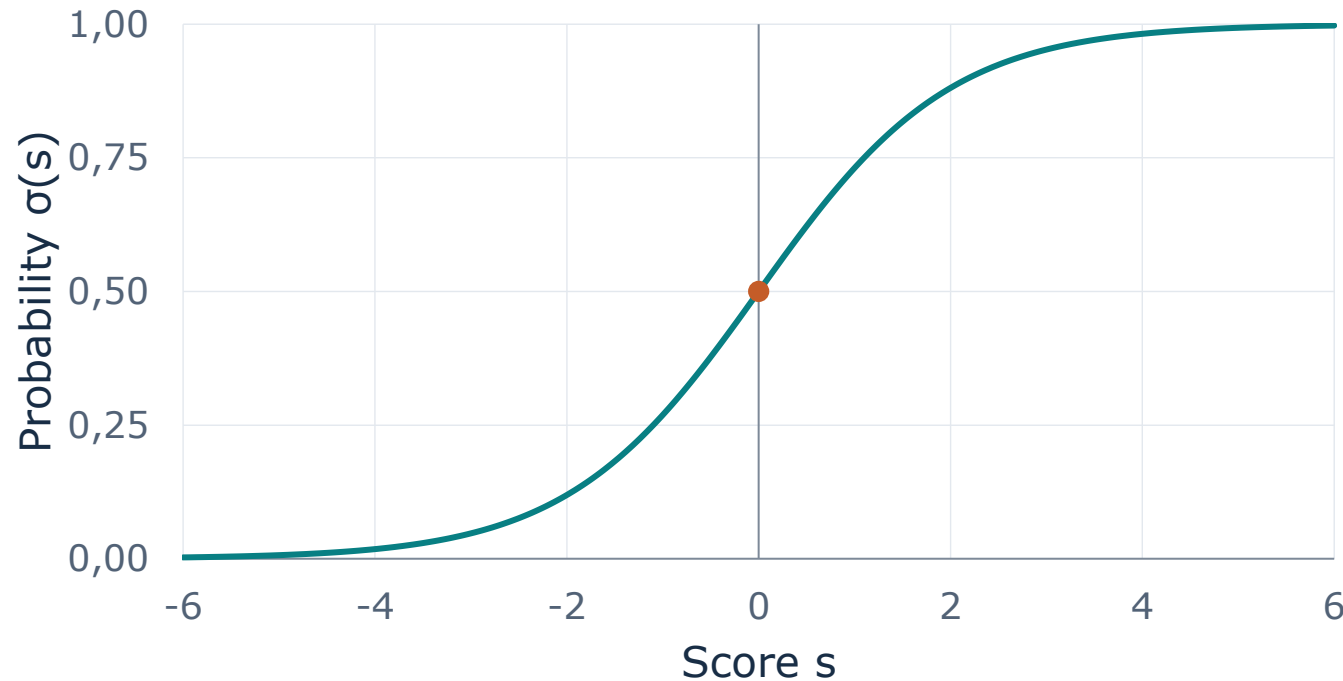
$$b \in \mathbb{R}, \quad \mathbf{w} = (w_1, \dots, w_d)^{\top} \in \mathbb{R}^{d \times 1}$$

$$\boldsymbol{\theta} = \begin{pmatrix} b \\ \mathbf{w} \end{pmatrix} = (b, w_1, \dots, w_d)^{\top} \in \mathbb{R}^{(d+1) \times 1}$$

However, $f_{\boldsymbol{\theta}}(\mathbf{x})$ can be **any real number**!
Hence, it cannot serve directly as a probability.
Rather, $f_{\boldsymbol{\theta}}(\mathbf{x})$ is a **score** (also called a **logit**)
that we have to turn into a probability.

The sigmoid function

A smooth, increasing function turns each real score into a number between zero and one.



$$\sigma(s) := \frac{1}{1 + e^{-s}} = \frac{e^s}{1 + e^s}$$

$$0 < \sigma(s) < 1$$

$$\sigma(0) = \frac{1}{2}$$

Probabilities, labels and the boundary

Using the sigmoid σ we can turn scores into probabilities:

$$p_{\theta}(\mathbf{x}) := \sigma(f_{\theta}(\mathbf{x})) = \sigma(b + \mathbf{xw})$$

The corresponding **plug-in classifier** $h_{p_{\theta}}$ satisfies

$$h_{p_{\theta}}(\mathbf{x}) = 1 \quad \Leftrightarrow \quad p_{\theta}(\mathbf{x}) > \frac{1}{2} \quad \Leftrightarrow \quad f_{\theta}(\mathbf{x}) > 0 \quad \Leftrightarrow \quad b + \mathbf{xw} > 0$$

Decision boundary: the set of $\mathbf{x} \in \mathbb{R}^d$ such that $b + \mathbf{xw} = 0$

The **decision boundary** is a hyperplane with d features,
provided the weight vector $\mathbf{w}$ is nonzero.

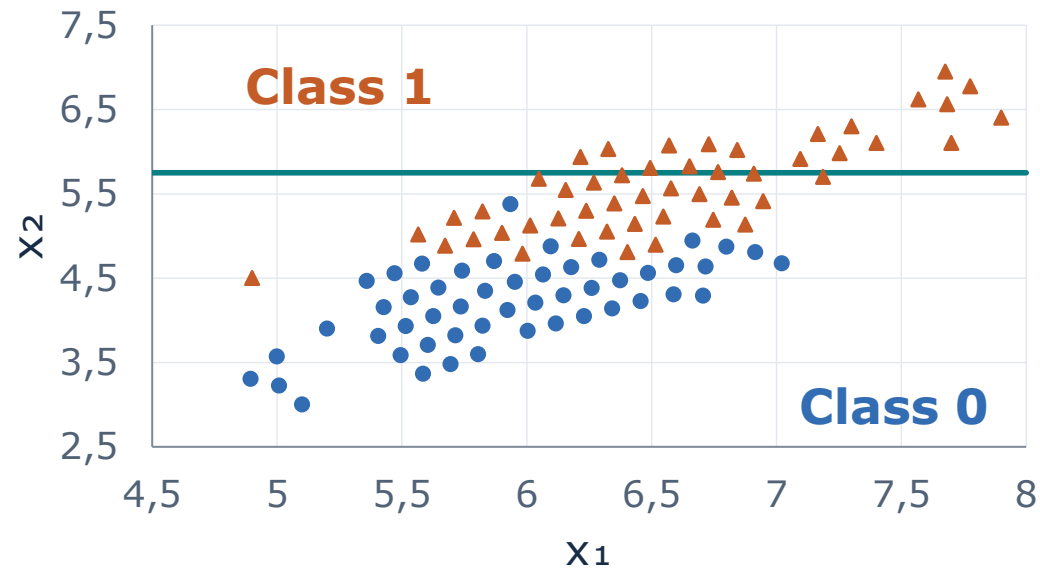
If $d = 1$, $b + x_1 w_1 = 0$ determines a point of the Euclidean line

If $d = 2$, $b + x_1 w_1 + x_2 w_2 = 0$ determines a straight line of the Euclidean plane

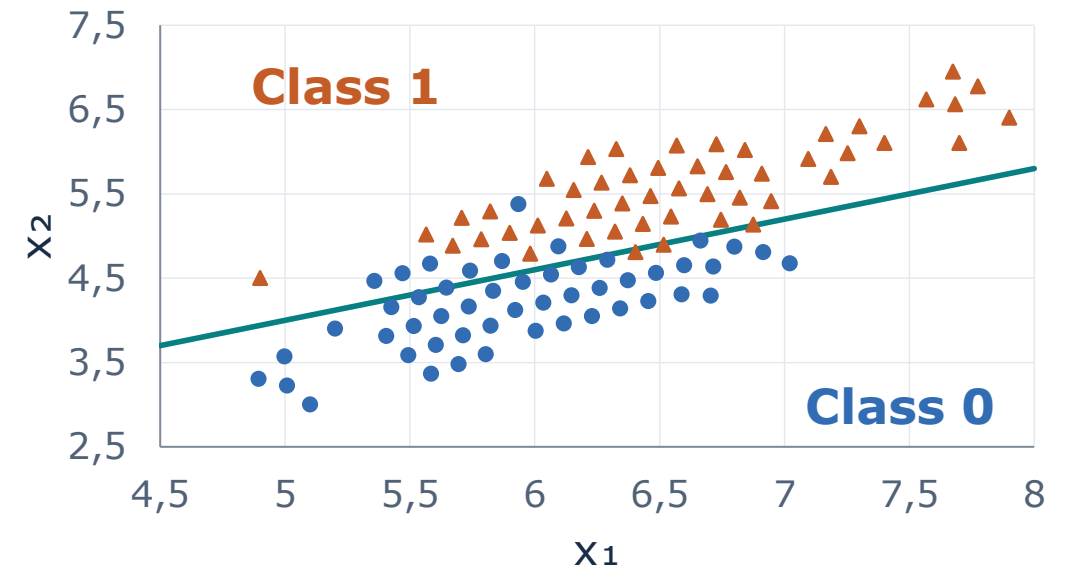
If $d = 3$, $b + x_1 w_1 + x_2 w_2 + x_3 w_3 = 0$ determines a plane of the Euclidean space

Comparing two candidate boundaries

Which of these two classifiers is better?
Count the number of mistakes!



• Class 0 ▲ Class 1 — Decision boundary



• Class 0 ▲ Class 1 — Decision boundary

$$\theta = (-5.75, 0, 1)^T, \quad x_2 = 5.75 \quad \theta = (-1, -0.6, 1)^T, \quad x_2 = 0.6x_1 + 1$$

The classifier on the right makes fewer mistakes on these observations

Today's lecture

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Gradient descent

More than two classes

Finding the parameters

We want the **smallest mistake count**.

Finding the parameter θ^* that minimizes the empirical 0-1 loss:

$$\theta \mapsto \mathcal{E}(\theta) = \sum_{i=1}^N \ell_{0,1}(h_{p_\theta}(\mathbf{x}_i), y_i)$$

is a **combinatorial** problem.

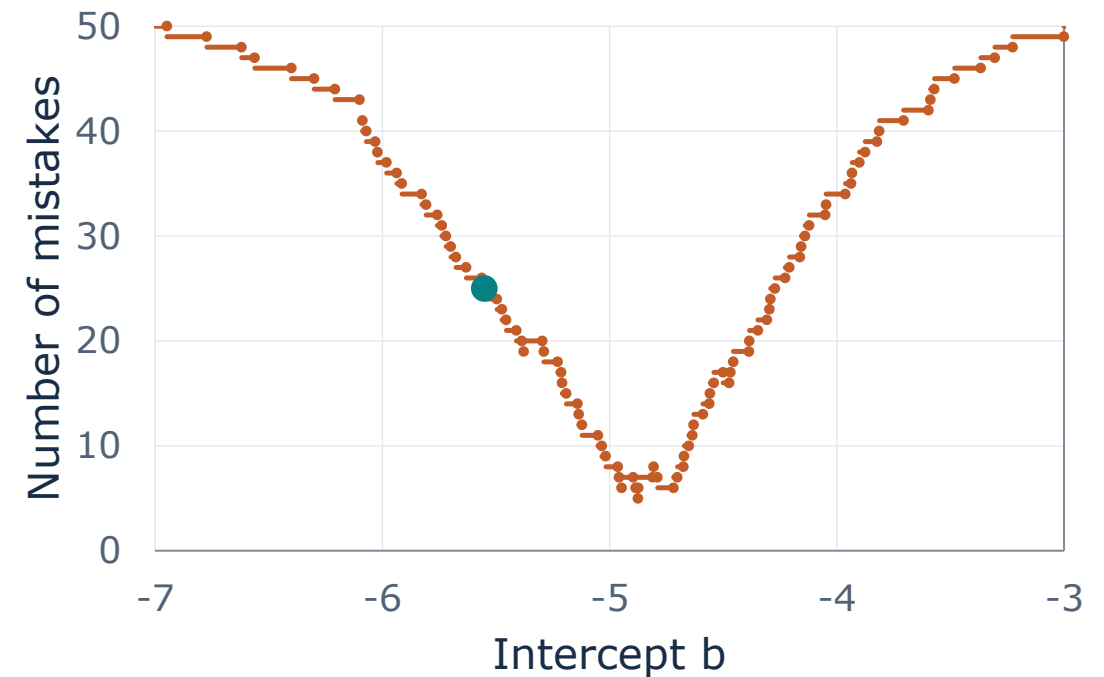
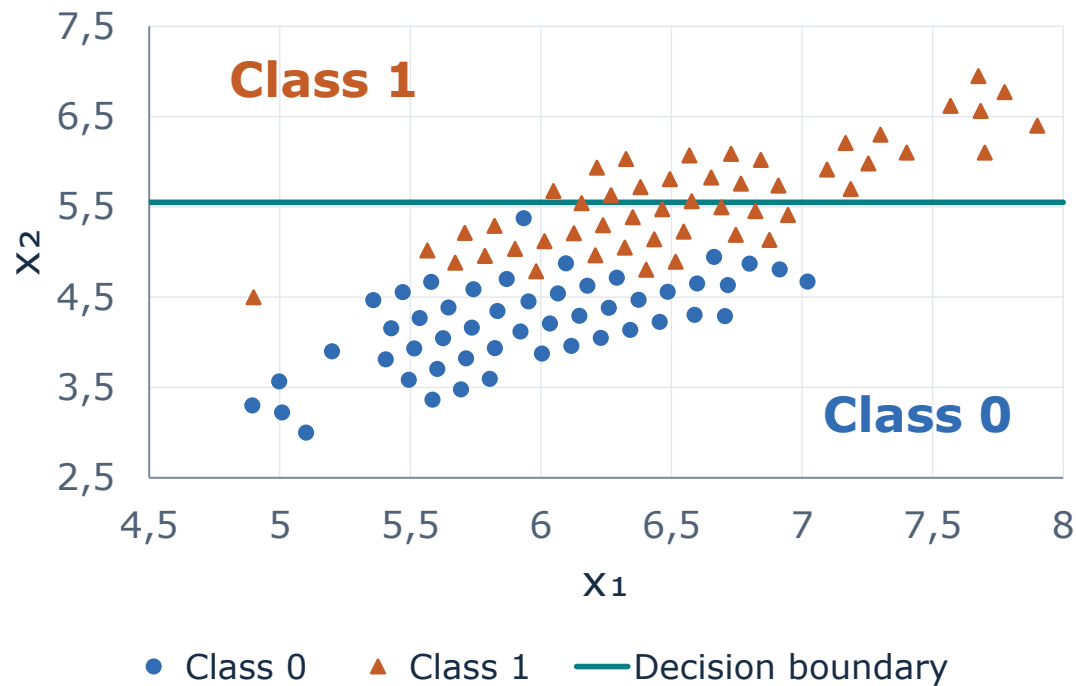
Can we compute an **approximate solution** via **gradient descent** as in regression?

$$\theta^{(t+1)} = \theta^{(t)} - \eta \nabla \mathcal{E}(\theta^{(t)}), \quad \eta > 0$$

What happens to the mistake count when we move the boundary a little?

Moving the boundary and counting mistakes

Fix the weights and vary only the intercept. The line moves parallel to itself.



$$b = -5.55, \quad w_1 = 0, \quad w_2 = 1$$

The count is **locally constant** and changes only when the line crosses an observed point.

Flat regions give no direction

If every observed score is **nonzero**
small parameter changes preserve **the signs** of the scores
and hence **all predicted labels stay the same**.

$$\mathcal{E}(\boldsymbol{\theta} + \Delta) = \mathcal{E}(\boldsymbol{\theta}) \quad \text{for sufficiently small } \Delta$$

The **gradient** of a **locally constant** function is **zero**!

$$\nabla \mathcal{E}(\boldsymbol{\theta}) = \mathbf{0}$$

Hence

$$\boldsymbol{\theta}^{(t+1)} = \boldsymbol{\theta}^{(t)} - \eta \nabla \mathcal{E}(\boldsymbol{\theta}^{(t)}) = \boldsymbol{\theta}^{(t)}$$

Gradient descent will **not move!!!**

What does the mistake count miss?

Suppose the **observed class is 0**. Both predictions below choose class 1.

	Prediction A	Prediction B
Probability of class 1	0.51	0.90
Predicted class	1	1
Zero-one loss	1	1
Probability of the observed class	0.49	0.10

At 0.51, the model is almost guessing: we want a milder penalty.

At 0.90, it is confidently wrong: we want a larger penalty.

Rather than scoring only the predicted label,
we **penalise assigning low probability to the observed class**.

Logistic loss: a surrogate for fitting probabilities

For $0 < p < 1$ and $y \in \{0, 1\}$, take the **negative logarithm** of the probability assigned to the observed class.

$$\ell_{\log}(p, y) = -y\log(p) - (1 - y)\log(1 - p) = \begin{cases} -\log(p), & y = 1, \\ -\log(1 - p), & y = 0. \end{cases}$$

The **logistic loss** makes us pay the negative logarithm of the probability we assign to the correct label.

Wrong, but uncertain

$$\begin{aligned} \ell_{\log}(0.51, 0) &= -\log(0.49) \\ &\approx 0.713 \end{aligned}$$

Confidently wrong

$$\begin{aligned} \ell_{\log}(0.90, 0) &= -\log(0.10) \\ &\approx 2.303 \end{aligned}$$

Logistic loss is also called **binary cross-entropy**. We use natural logarithms.

The empirical logistic loss

The observations stay fixed.

The **probability predictions** depend on the parameters.

$$p_{\theta}(\mathbf{x}_i) = \sigma(b + \mathbf{x}_i \mathbf{w})$$

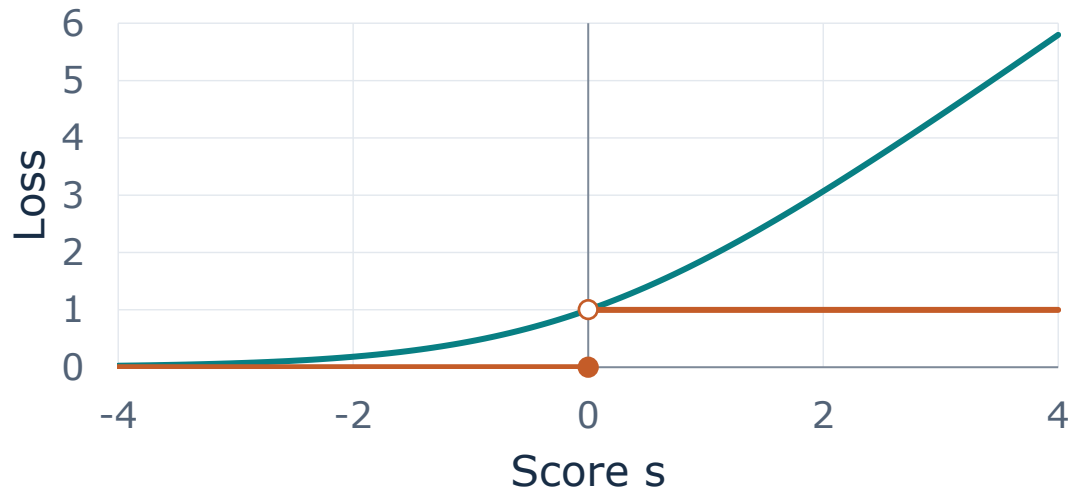
$$\mathcal{L}_{\log}(\theta) = \sum_{i=1}^N \ell_{\log}(p_{\theta}(\mathbf{x}_i), y_i)$$

We fit this **smooth surrogate loss using **gradient descent**, while the mistake count $\varepsilon(\theta)$ remains our classification measure.**

Comparing the two losses

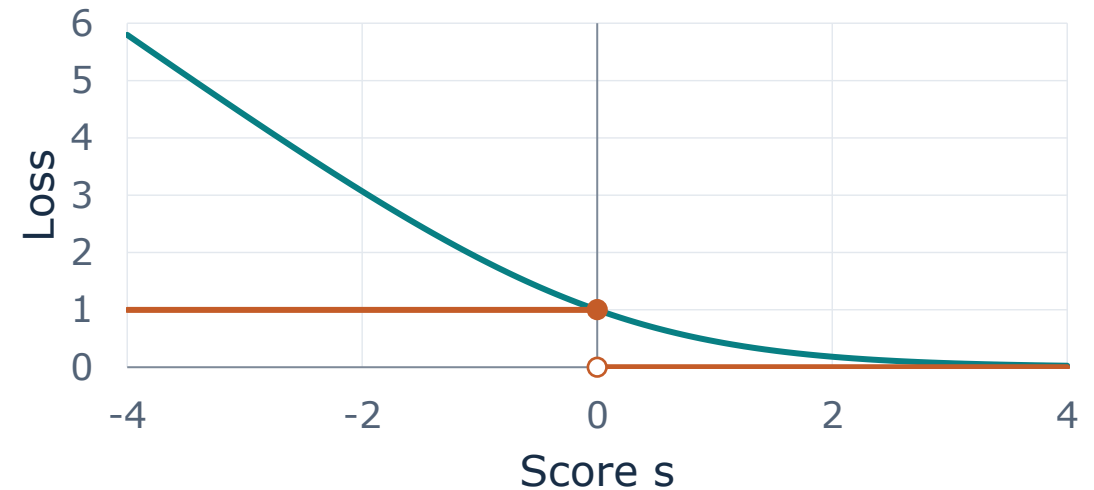
A mistake gives the observed class probability at most $1/2$,
so its logistic loss is at least $-\log\left(\frac{1}{2}\right) = \log(2)$.

Observed label $y = 0$



— Logistic loss / $\log 2$ — Zero-one loss

Observed label $y = 1$



— Logistic loss / $\log 2$ — Zero-one loss

$$\ell_{0,1}\left(\text{step}\left(\sigma(s) - \frac{1}{2}\right), y\right) \leq \frac{\ell_{\log}(\sigma(s), y)}{\log(2)}$$

Bounding the number of mistakes

The inequality holds for each observation

We can therefore add it over the dataset

$$\begin{aligned}\mathcal{E}(\boldsymbol{\theta}) &= \sum_{i=1}^N \ell_{0,1}(h_{p_{\boldsymbol{\theta}}}(\mathbf{x}_i), y_i) \\ &= \sum_{i=1}^N \ell_{0,1}\left(\text{step}\left(\sigma(f_{\boldsymbol{\theta}}(\mathbf{x}_i))\right) - \frac{1}{2}, y_i\right) \leq \sum_{i=1}^N \frac{\ell_{\log}(\sigma(f_{\boldsymbol{\theta}}(\mathbf{x}_i)), y_i)}{\log(2)} \\ &= \frac{1}{\log(2)} \sum_{i=1}^N \ell_{\log}(p_{\boldsymbol{\theta}}(\mathbf{x}_i), y_i) = \frac{1}{\log(2)} \cdot \mathcal{L}_{\log}(\boldsymbol{\theta})\end{aligned}$$

A control over the empirical logistic loss
gives a control on the number of mistakes

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The design matrix and the parameters

As in Lecture 2, a first column of ones includes the intercept.

$$\mathbf{X} = \begin{pmatrix} 1 & x_{11} & \cdots & x_{1d} \\ \vdots & \vdots & & \vdots \\ 1 & x_{N1} & \cdots & x_{Nd} \end{pmatrix} \quad \boldsymbol{\theta} = \begin{pmatrix} b \\ w_1 \\ \vdots \\ w_d \end{pmatrix}$$

$$\mathbf{X} \in \mathbb{R}^{N \times (d+1)}, \quad \boldsymbol{\theta} \in \mathbb{R}^{(d+1) \times 1}$$

$$\theta_1 = b, \quad \theta_{k+1} = w_k, \quad k = 1, \dots, d$$

The first coordinate is the intercept. The remaining coordinates are the weights.

The three functions

Fix the i th input/output pair $(\mathbf{x}_i, y_i)$ and vary the model parameters $\boldsymbol{\theta}$.

Define affine score corresponding to the i th input $\mathbf{x}_i$

$$F_i(\boldsymbol{\theta}) := f_{\boldsymbol{\theta}}(\mathbf{x}_i) = b + \mathbf{x}_i \mathbf{w} = (1, \mathbf{x}_i) \boldsymbol{\theta} = \sum_{r=1}^{d+1} X_{ir} \theta_r$$

Recall the definition of the sigmoid function σ

$$\sigma(s) = \frac{1}{1 + e^{-s}} \quad (s \in \mathbb{R})$$

Define the logistic loss corresponding to the i th output y_i

$$G_i(p) := \ell_{\log}(p, y_i)$$

The chain rule for one observation

The loss for the i th observation is the **composition** of these three functions.

$$\ell_{\log}(p_{\boldsymbol{\theta}}(\mathbf{x}_i), y_i) = (G_i \circ \sigma \circ F_i)(\boldsymbol{\theta}) = G_i \left(\sigma(F_i(\boldsymbol{\theta})) \right)$$

To compute the partial derivatives with respect to θ_j we apply the **chain rule to this composition.**

$$\begin{aligned} \frac{\partial}{\partial \theta_j} \ell_{\log}(p_{\boldsymbol{\theta}}(\mathbf{x}_i), y_i) &= \frac{\partial}{\partial \theta_j} (G_i \circ \sigma \circ F_i)(\boldsymbol{\theta}) = G'_i \left(\sigma(F_i(\boldsymbol{\theta})) \right) \cdot \sigma'(F_i(\boldsymbol{\theta})) \cdot \frac{\partial F_i}{\partial \theta_j}(\boldsymbol{\theta}) \\ &= \frac{\partial \ell_{\log}}{\partial p} \left(\sigma(f_{\boldsymbol{\theta}}(\mathbf{x}_i)), y_i \right) \cdot \sigma'(f_{\boldsymbol{\theta}}(\mathbf{x}_i)) \cdot \frac{\partial f_{\boldsymbol{\theta}}(\mathbf{x}_i)}{\partial \theta_j} \\ &= \underbrace{\frac{\partial \ell_{\log}}{\partial p} (p_{\boldsymbol{\theta}}(\mathbf{x}_i), y_i)}_{\text{first term}} \cdot \underbrace{\sigma'(f_{\boldsymbol{\theta}}(\mathbf{x}_i))}_{\text{second term}} \cdot \underbrace{\frac{\partial f_{\boldsymbol{\theta}}(\mathbf{x}_i)}{\partial \theta_j}}_{\text{third term}} \end{aligned}$$

First term: Differentiating the logistic loss

Hold the observed label fixed and vary the probability.

$$\ell_{\log}(p, y) = -y\log(p) - (1 - y)\log(1 - p)$$

$$\begin{aligned}\frac{\partial \ell_{\log}}{\partial p}(p, y) &= -\frac{y}{p} + \frac{1 - y}{1 - p} = \frac{-y(1 - p) + (1 - y)p}{p(1 - p)} \\ &= \frac{-y + yp + p - yp}{p(1 - p)} = \frac{p - y}{p(1 - p)}\end{aligned}$$

$$\boxed{\frac{\partial \ell_{\log}}{\partial p}(p_{\theta}(\mathbf{x}_i), y_i) = \frac{p_{\theta}(\mathbf{x}_i) - y_i}{p_{\theta}(\mathbf{x}_i)(1 - p_{\theta}(\mathbf{x}_i))}}$$

Second term: Differentiating the sigmoid

Write the **sigmoid as a composition** and apply the **chain rule** again.

Inner function u

$$u(s) = 1 + e^{-s}, \quad u'(s) = -e^{-s}$$

Outer function v

$$v(z) = \frac{1}{z}, \quad v'(z) = -\frac{1}{z^2}$$

$$\sigma(s) = \frac{1}{1 + e^{-s}} = v(u(s))$$

$$\sigma'(s) = v'(u(s)) u'(s) = -\frac{1}{(1 + e^{-s})^2} (-e^{-s}) = \frac{e^{-s}}{(1 + e^{-s})^2}$$

$$= \frac{1}{1 + e^{-s}} \cdot \frac{e^{-s}}{1 + e^{-s}} = \frac{1}{1 + e^{-s}} \cdot \left(1 - \frac{1}{1 + e^{-s}}\right) = \sigma(s)(1 - \sigma(s))$$

$$\boxed{\sigma'(f_{\theta}(\mathbf{x}_i)) = \sigma(f_{\theta}(\mathbf{x}_i)) (1 - \sigma(f_{\theta}(\mathbf{x}_i))) = p_{\theta}(\mathbf{x}_i)(1 - p_{\theta}(\mathbf{x}_i))}$$

Third term: Differentiating the affine score

The inputs are fixed, so only the matching parameter contributes.

The affine score

$$f_{\boldsymbol{\theta}}(\mathbf{x}_i) = \sum_{r=1}^{d+1} X_{ir} \theta_r$$

The Kronecker delta

$$\frac{\partial \theta_r}{\partial \theta_j} = \delta_{rj} = \begin{cases} 1, & r = j, \\ 0, & r \neq j. \end{cases}$$

$$\frac{\partial f_{\boldsymbol{\theta}}(\mathbf{x}_i)}{\partial \theta_j} = \frac{\partial}{\partial \theta_j} \sum_{r=1}^{d+1} X_{ir} \theta_r = \sum_{r=1}^{d+1} X_{ir} \frac{\partial \theta_r}{\partial \theta_j} = \sum_{r=1}^{d+1} X_{ir} \delta_{rj} = X_{ij}$$

$$\boxed{\frac{\partial f_{\boldsymbol{\theta}}(\mathbf{x}_i)}{\partial \theta_j} = X_{ij}} \quad j = 1, \dots, d + 1$$

Putting the three terms together

Substitute the first, second and third terms into the chain rule.

$$\begin{aligned} & \frac{\partial}{\partial \theta_j} \ell_{\log}(p_{\theta}(\mathbf{x}_i), y_i) \\ &= \underbrace{\frac{\partial \ell_{\log}}{\partial p}(p_{\theta}(\mathbf{x}_i), y_i)}_{\text{first term}} \cdot \underbrace{\sigma'(f_{\theta}(\mathbf{x}_i))}_{\text{second term}} \cdot \underbrace{\frac{\partial f_{\theta}(\mathbf{x}_i)}{\partial \theta_j}}_{\text{third term}} \\ &= \underbrace{\frac{p_{\theta}(\mathbf{x}_i) - y_i}{p_{\theta}(\mathbf{x}_i)(1 - p_{\theta}(\mathbf{x}_i))}}_{\text{first term}} \cdot \underbrace{p_{\theta}(\mathbf{x}_i)(1 - p_{\theta}(\mathbf{x}_i))}_{\text{second term}} \cdot \underbrace{X_{ij}}_{\text{third term}} \\ &= (p_{\theta}(\mathbf{x}_i) - y_i)X_{ij} \end{aligned}$$

Empirical gradient and gradient descent

The empirical loss is a sum, so we add the observation derivatives.

$$\frac{\partial \mathcal{L}_{\log}}{\partial \theta_j}(\boldsymbol{\theta}) = \sum_{i=1}^N \frac{\partial}{\partial \theta_j} \ell_{\log}(p_{\boldsymbol{\theta}}(\mathbf{x}_i), y_i) = \sum_{i=1}^N X_{ij}(p_{\boldsymbol{\theta}}(\mathbf{x}_i) - y_i)$$

Sigmoid applied to each entry

$$\boldsymbol{\sigma}((s_1, \dots, s_N)^\top) := (\sigma(s_1), \dots, \sigma(s_N))^\top$$

One probability per observation

$$\mathbf{p}(\boldsymbol{\theta}) := (p_{\boldsymbol{\theta}}(\mathbf{x}_1), \dots, p_{\boldsymbol{\theta}}(\mathbf{x}_N))^\top$$

$$\mathbf{p}(\boldsymbol{\theta}) = \boldsymbol{\sigma}(\mathbf{X}\boldsymbol{\theta}), \quad \mathbf{y} = (y_1, \dots, y_N)^\top$$

Gradient

$$\nabla \mathcal{L}_{\log}(\boldsymbol{\theta}) = \mathbf{X}^\top(\mathbf{p}(\boldsymbol{\theta}) - \mathbf{y}) = \mathbf{X}^\top(\boldsymbol{\sigma}(\mathbf{X}\boldsymbol{\theta}) - \mathbf{y})$$

Fitting logistic regression

$$\boldsymbol{\theta}^{(t+1)} = \boldsymbol{\theta}^{(t)} - \eta \nabla \mathcal{L}_{\log}(\boldsymbol{\theta}^{(t)}) = \boldsymbol{\theta}^{(t)} - \eta \mathbf{X}^\top(\boldsymbol{\sigma}(\mathbf{X}\boldsymbol{\theta}^{(t)}) - \mathbf{y})$$

Stationary points and minimisers

Fact: the empirical logistic loss is **convex** in the parameters.
Therefore, **any stationary point is a global minimiser.**

However: a finite minimiser need not exist!!!
See the notes for a counterexample.

Also: Even when a finite minimiser exists,
the stationarity equations form a **nonlinear system**,
generally with **no closed-form solution**.

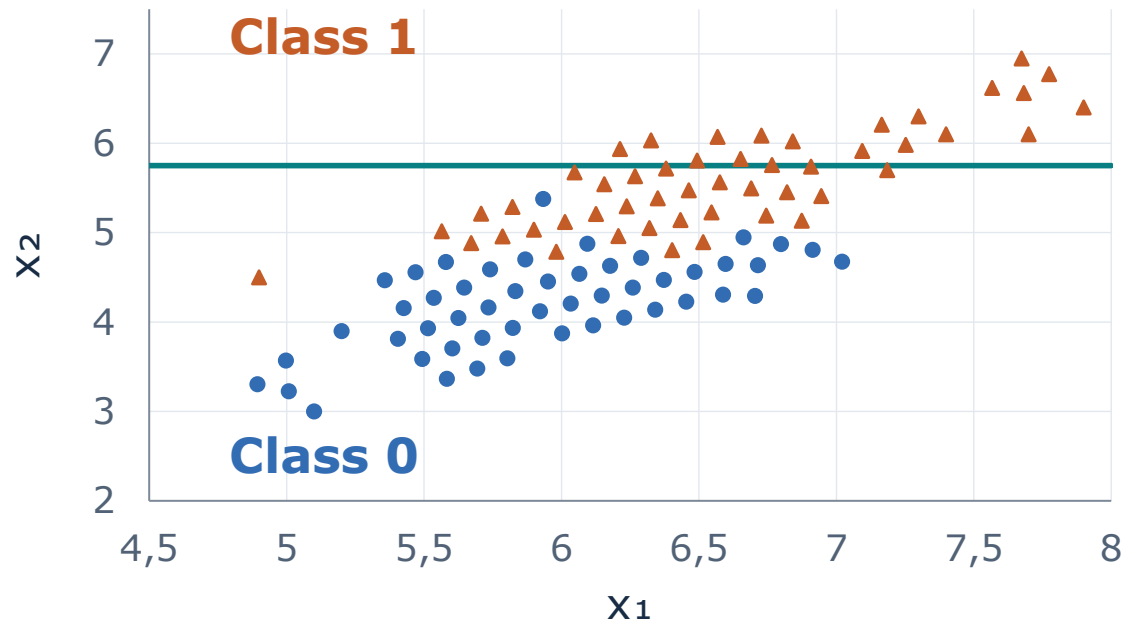
Warning 1: decreasing the empirical logistic loss
does not guarantee fewer mistakes (a lower empirical 0-1 loss).

Warning 2: gradient descent still needs an appropriate learning rate.
A step that is too large can increase the loss.

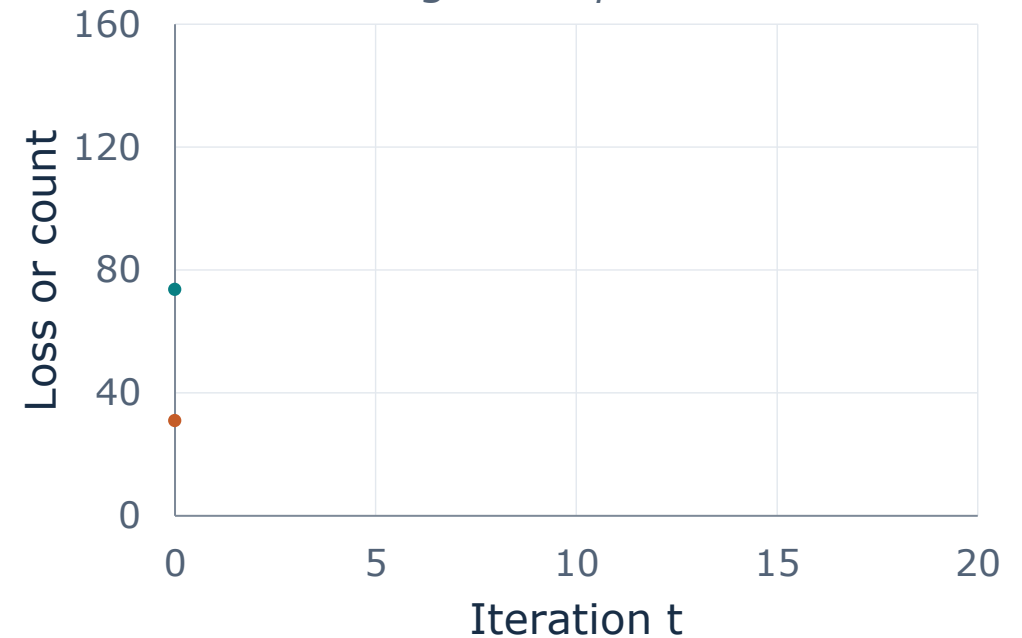
Gradient descent: $\eta = 0.001, t = 0$

Initial coefficients: $b^{(0)} = -5.75,$ $w_1^{(0)} = 0,$ $w_2^{(0)} = 1$

Learning rate: $\eta = 0.001$



● Class 0 ▲ Class 1 — Decision boundary



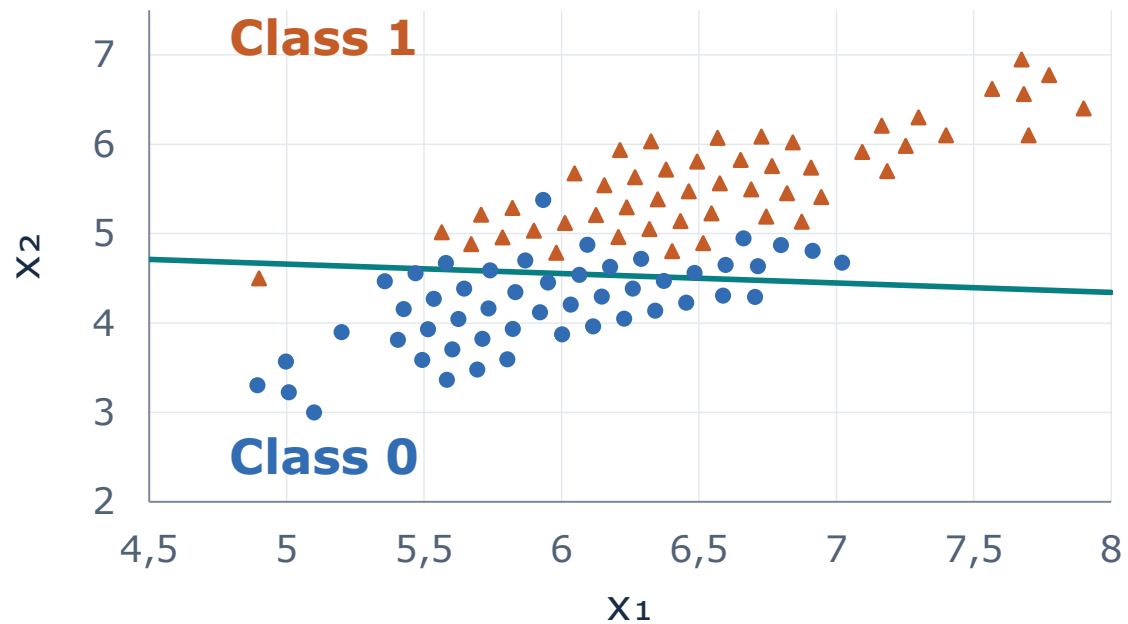
—●— Logistic loss / log 2 —●— Number of mistakes

$$x_2 = 5.7500 + 0.0000x_1$$

Logistic loss: 51.048
Mistakes: 31 of 100

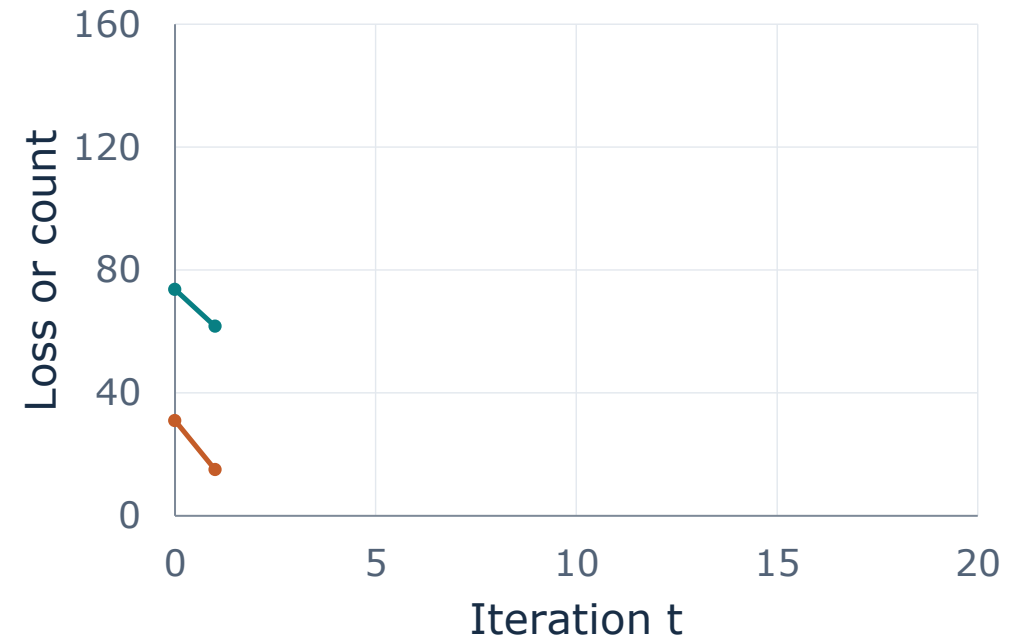
Gradient descent: $\eta = 0.001, t = 1$

Coefficients: $b^{(1)} \approx -5.732582$ $w_1^{(1)} \approx 0.116619$ $w_2^{(1)} \approx 1.105206$



● Class 0 ▲ Class 1 — Decision boundary

$$x_2 = 5.1869 - 0.1055x_1$$

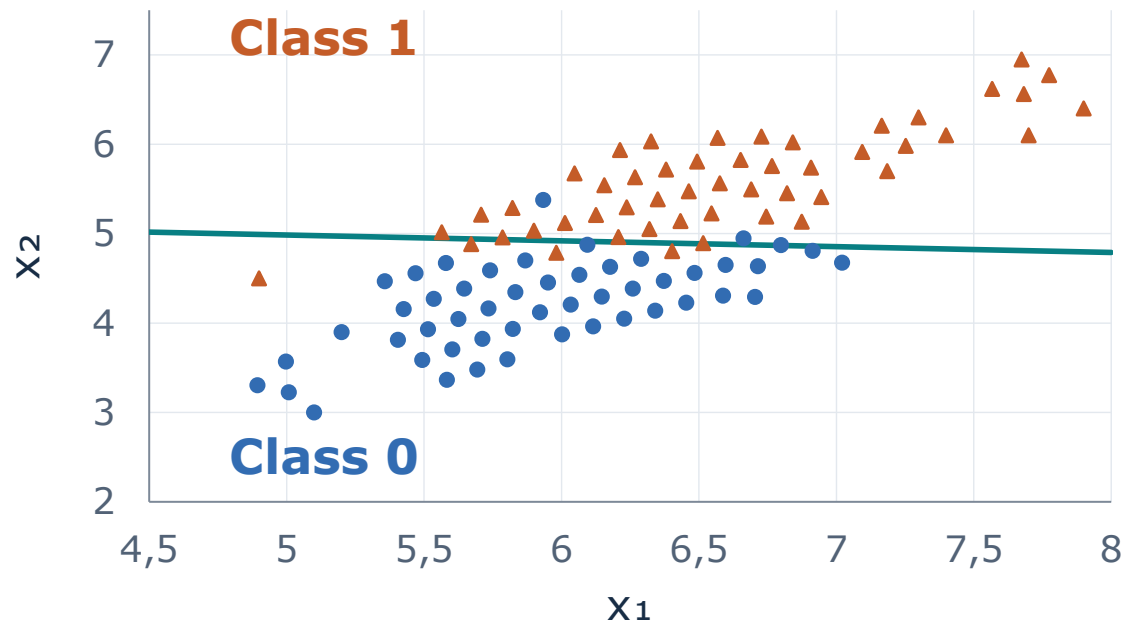


—●— Logistic loss / log 2 —●— Number of mistakes

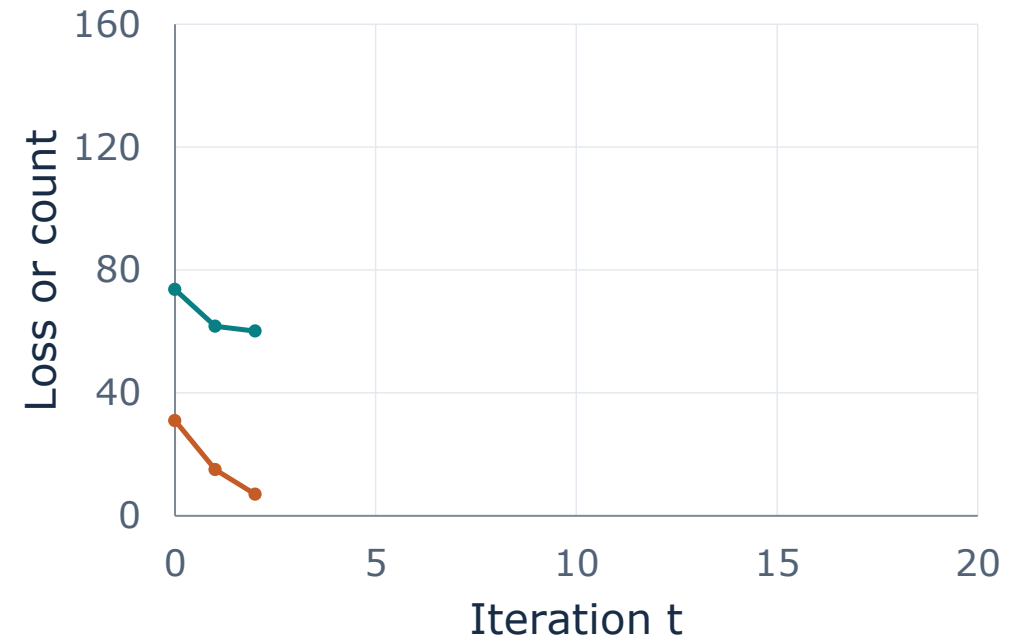
Logistic loss: 42.733
Mistakes: 15 of 100

Gradient descent: $\eta = 0.001, t = 2$

Coefficients: $b^{(2)} \approx -5.740878$ $w_1^{(2)} \approx 0.070300$ $w_2^{(2)} \approx 1.081132$



● Class 0 ▲ Class 1 — Decision boundary



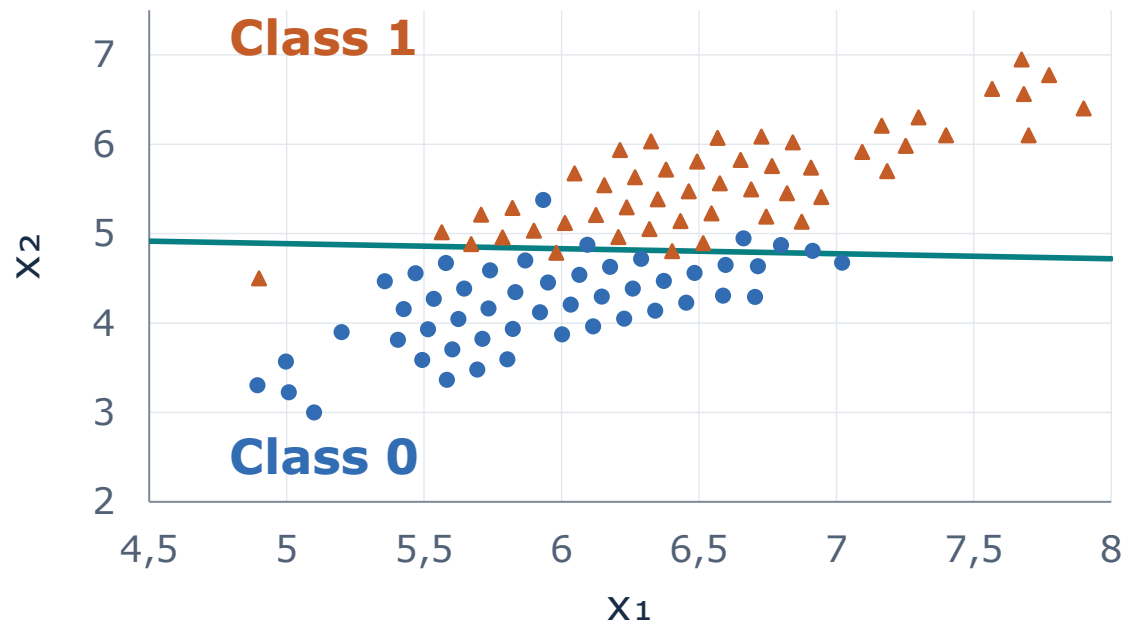
—●— Logistic loss / log 2 —●— Number of mistakes

$$x_2 = 5.3101 - 0.0650x_1$$

Logistic loss: 41.668
Mistakes: 7 of 100

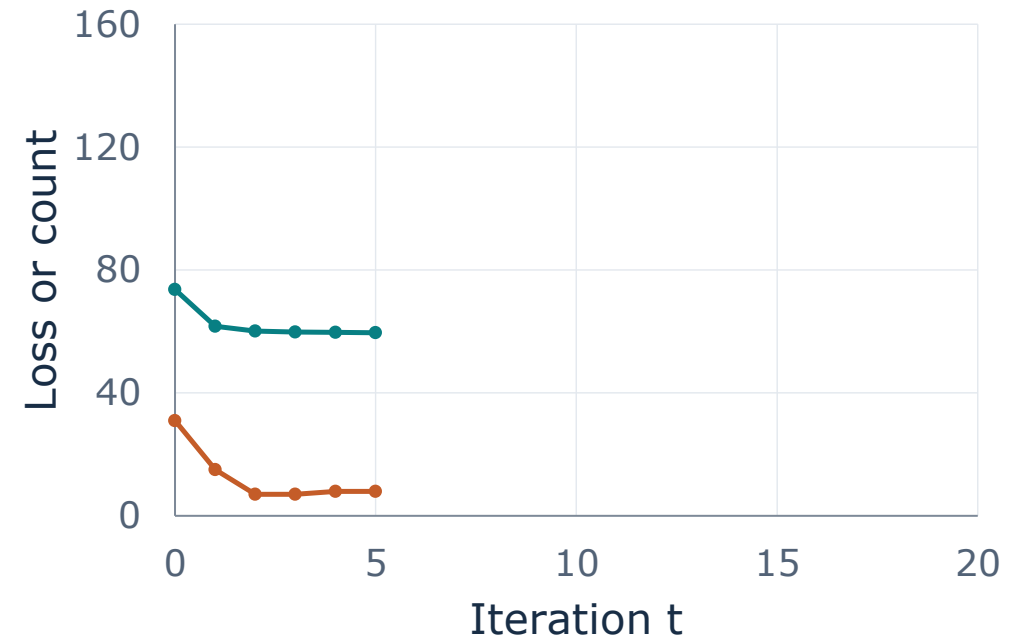
Gradient descent: $\eta = 0.001, t = 5$

Coefficients: $b^{(5)} \approx -5.744795$ $w_1^{(5)} \approx 0.062560$ $w_2^{(5)} \approx 1.111266$



● Class 0 ▲ Class 1 — Decision boundary

$$x_2 = 5.1696 - 0.0563x_1$$

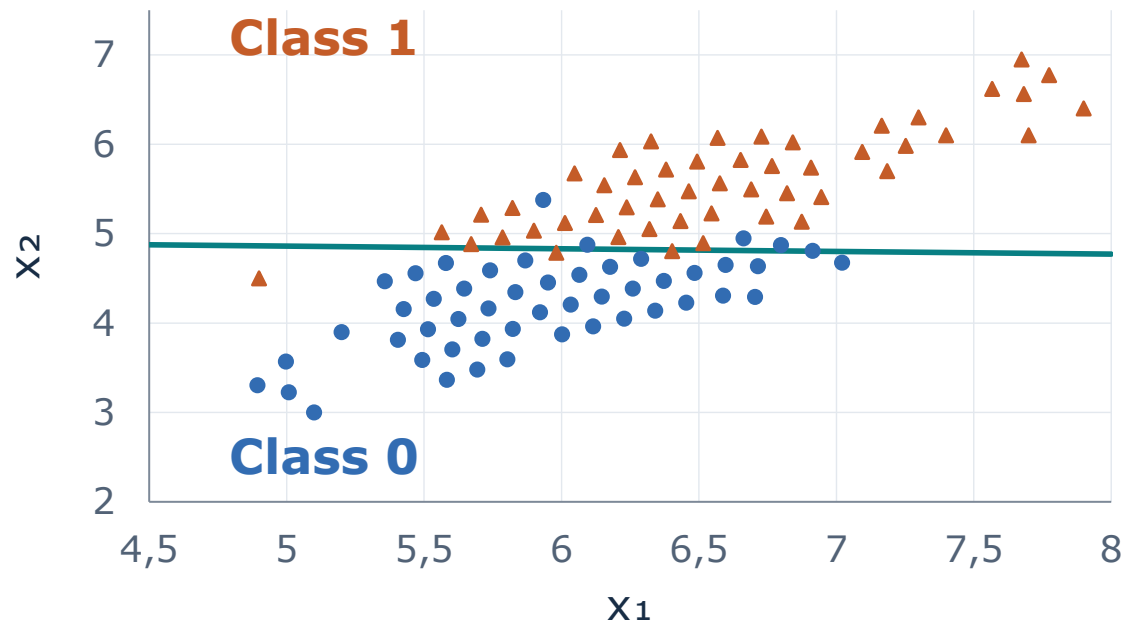


— Logistic loss / log 2 — Number of mistakes

Logistic loss: 41.300
Mistakes: 8 of 100

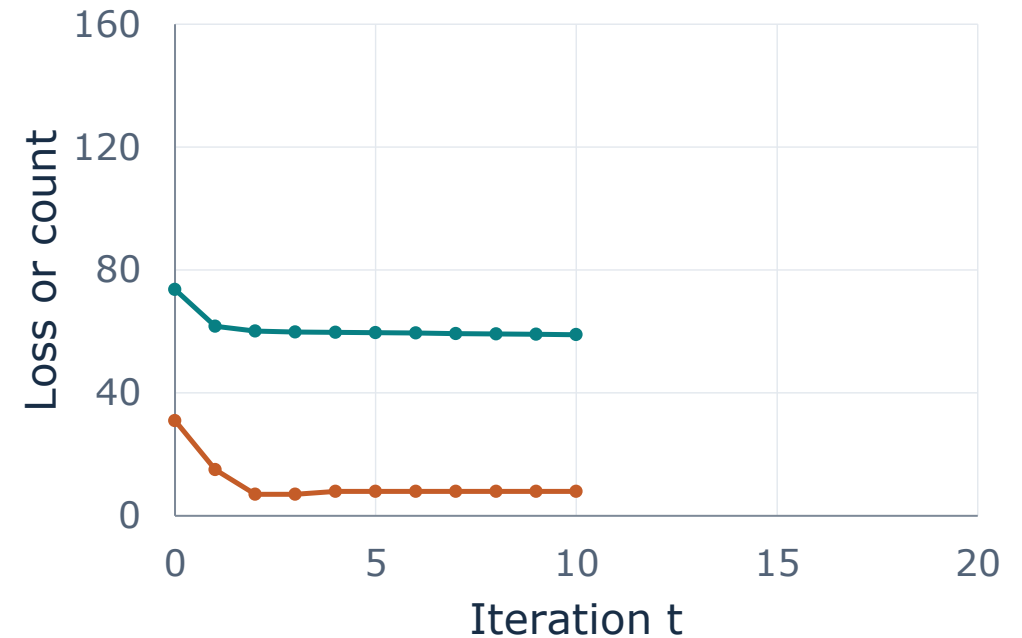
Gradient descent: $\eta = 0.001, t = 10$

Coefficients: $b^{(10)} \approx -5.753743$ $w_1^{(10)} \approx 0.034272$ $w_2^{(10)} \approx 1.148470$



● Class 0 ▲ Class 1 — Decision boundary

$$x_2 = 5.0099 - 0.0298x_1$$

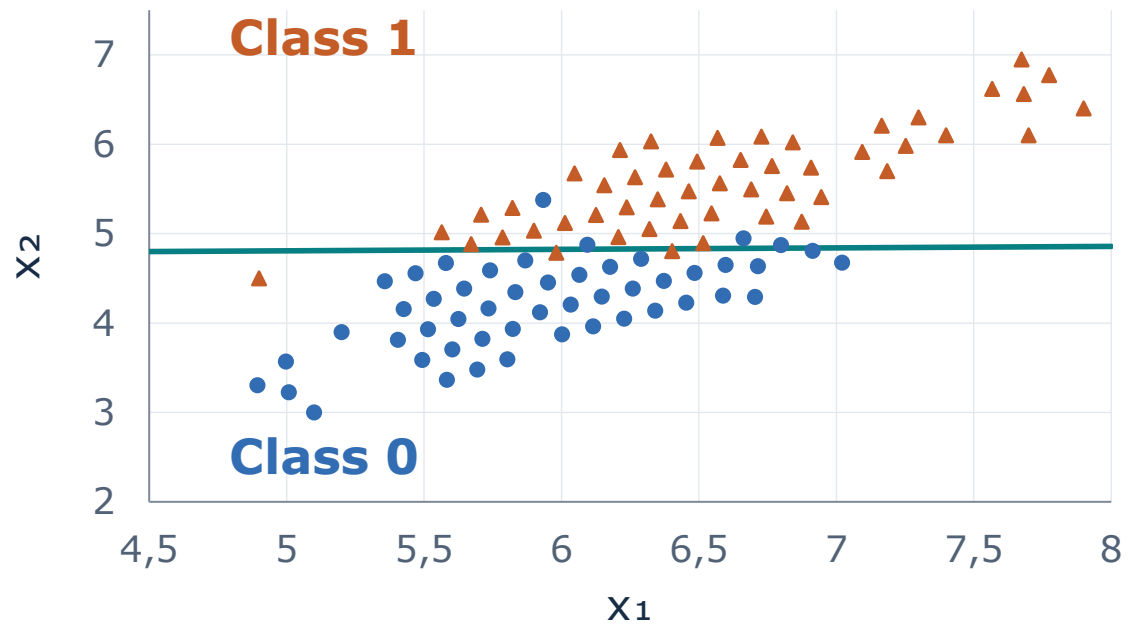


— Logistic loss / log 2 — Number of mistakes

Logistic loss: 40.848
Mistakes: 8 of 100

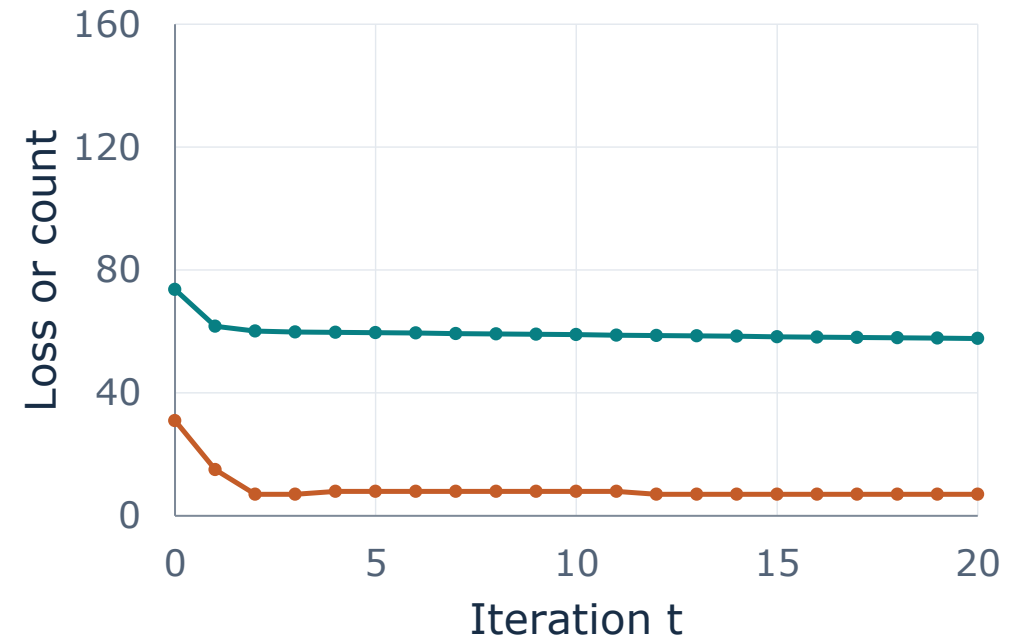
Gradient descent: $\eta = 0.001, t = 20$

Coefficients: $b^{(20)} \approx -5.771210$ $w_1^{(20)} \approx -0.020164$ $w_2^{(20)} \approx 1.221302$



● Class 0 ▲ Class 1 — Decision boundary

$$x_2 = 4.7255 + 0.0165x_1$$

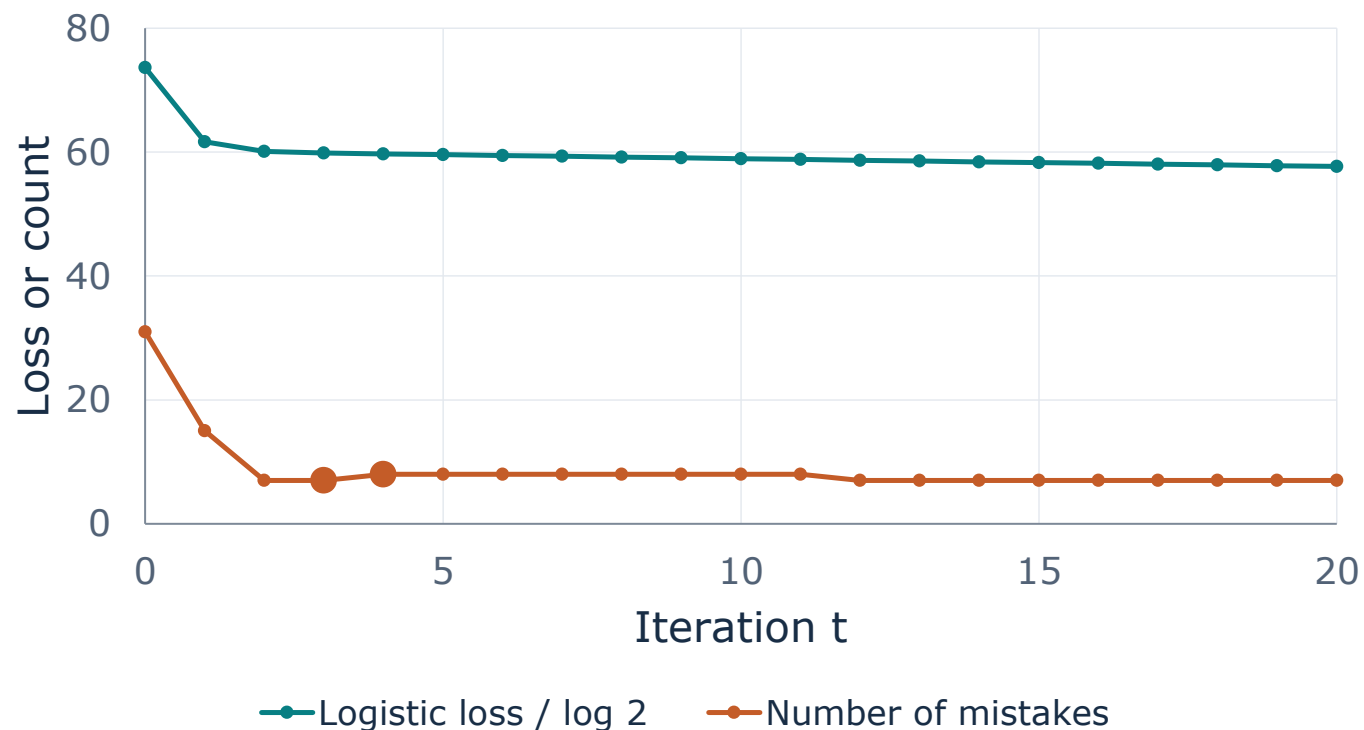


—●— Logistic loss / log 2 —●— Number of mistakes

Logistic loss: 39.992
Mistakes: 7 of 100

The loss can fall while mistakes increase

The same planar run shows the limitation of the surrogate.



Step	Logistic loss	Errors
3	41.4948	7
4	41.3932	8

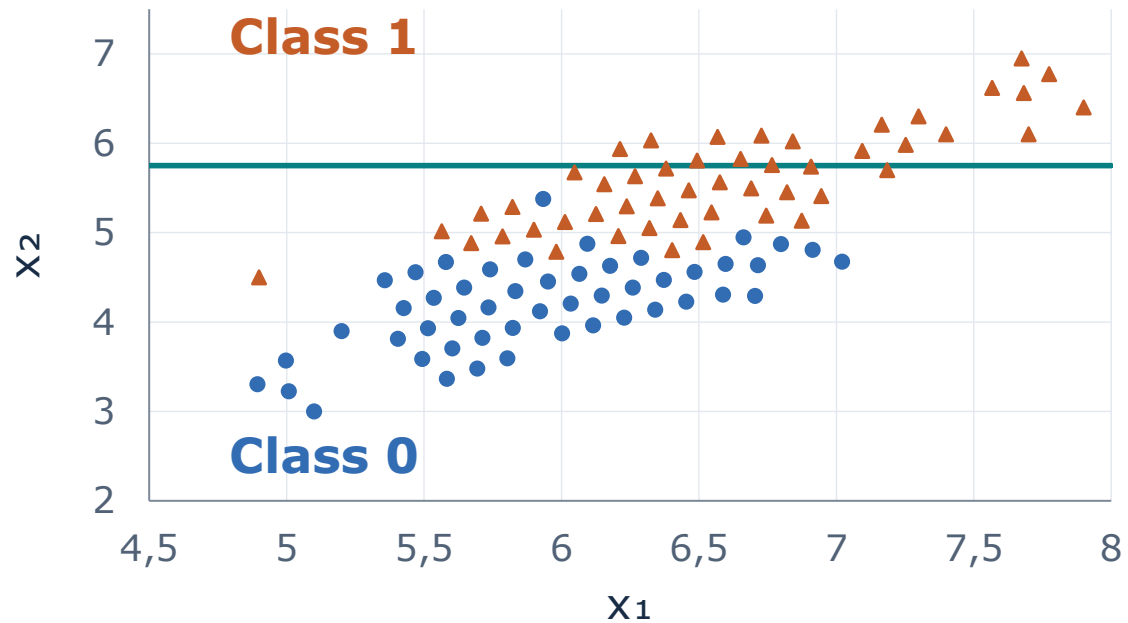
Of course: the bound still holds at both parameter choices.

Logistic loss can decrease even when the mistake count increases.

Gradient descent: $\eta = 0.002, t = 0$

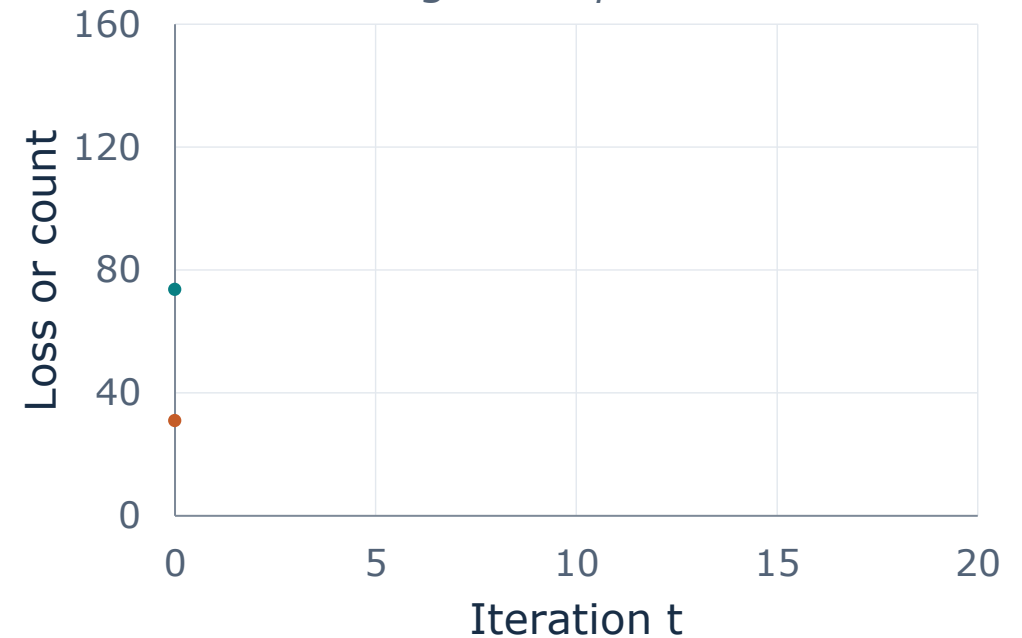
Initial coefficients: $b^{(0)} = -5.75,$ $w_1^{(0)} = 0,$ $w_2^{(0)} = 1$

Learning rate: $\eta = 0.002$



● Class 0 ▲ Class 1 — Decision boundary

$$x_2 = 5.7500 + 0.0000x_1$$

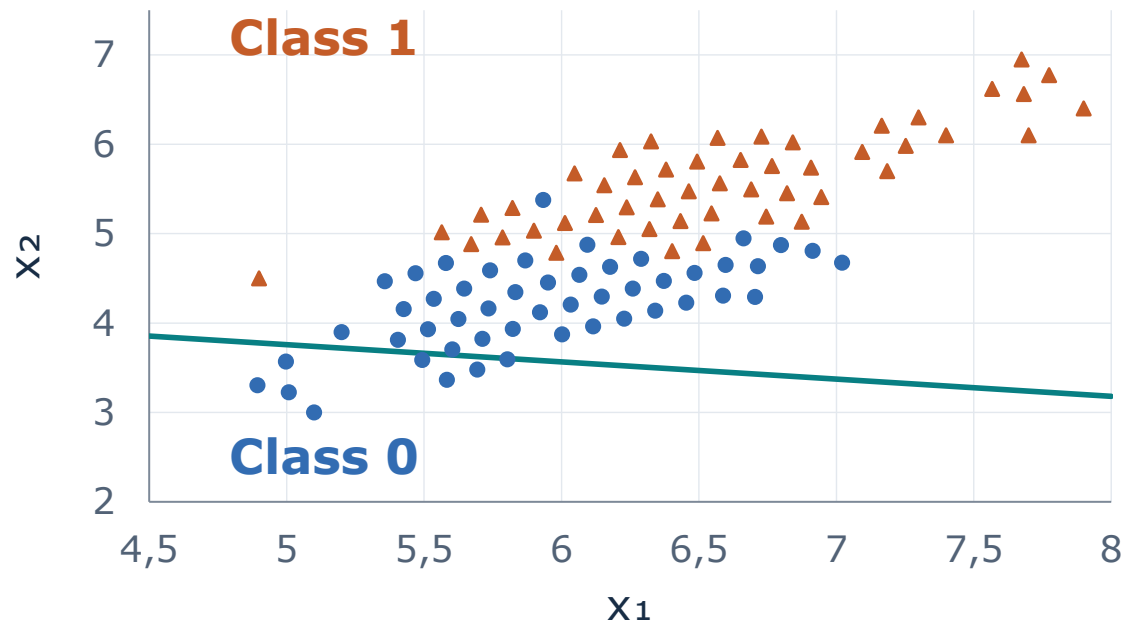


—●— Logistic loss / log 2 —●— Number of mistakes

Logistic loss: 51.048
Mistakes: 31 of 100

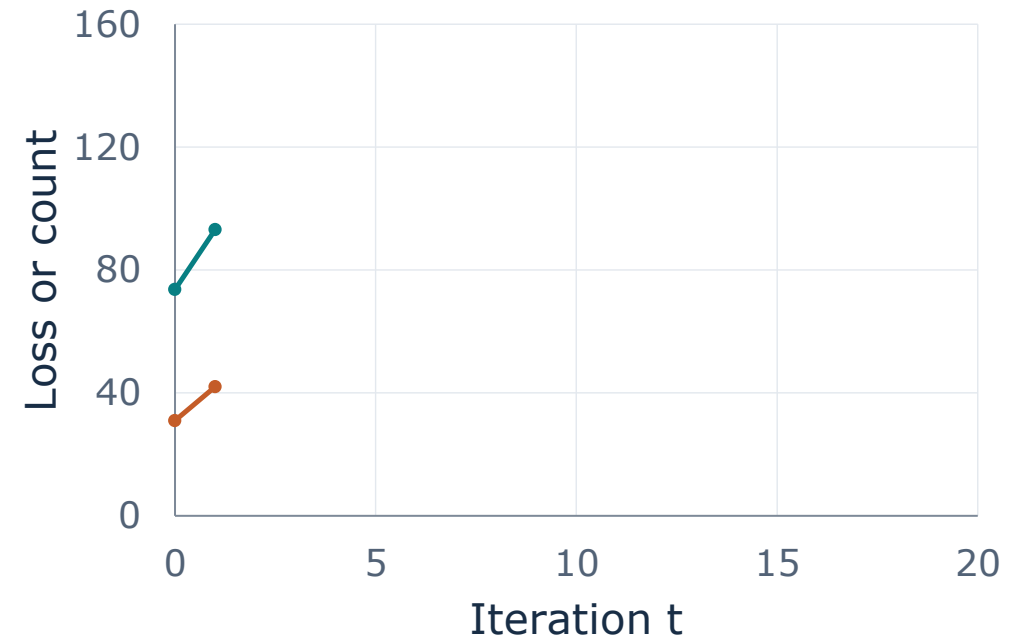
Gradient descent: $\eta = 0.002, t = 1$

Coefficients: $b^{(1)} \approx -5.715164$ $w_1^{(1)} \approx 0.233238$ $w_2^{(1)} \approx 1.210411$



● Class 0 ▲ Class 1 — Decision boundary

$$x_2 = 4.7217 - 0.1927x_1$$

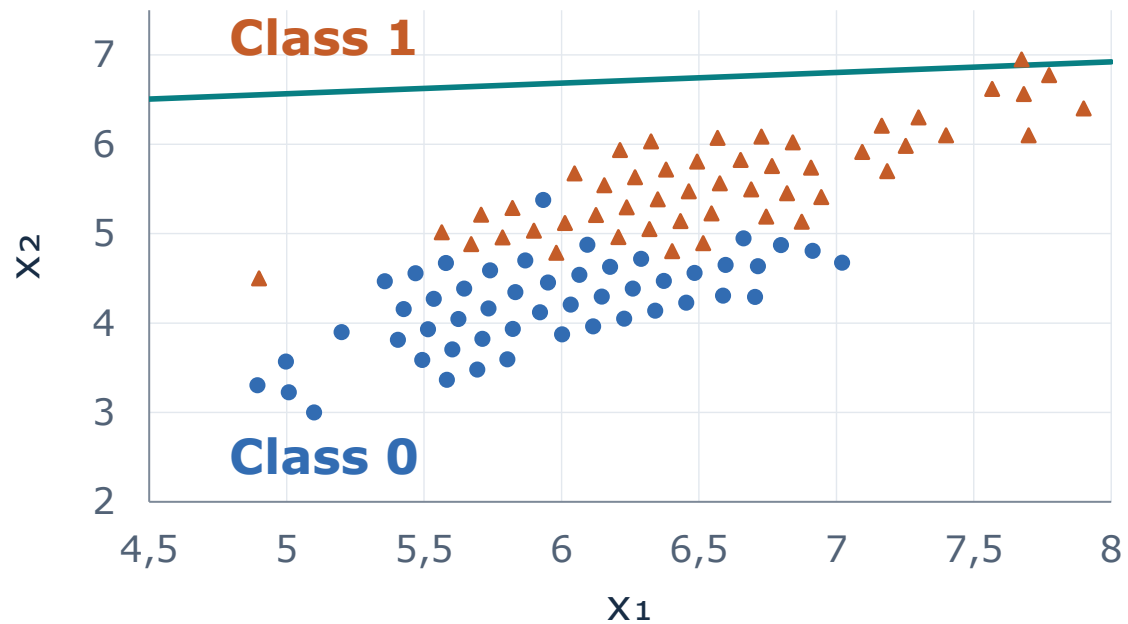


— Logistic loss / log 2 — Number of mistakes

Logistic loss: 64.556
Mistakes: 42 of 100

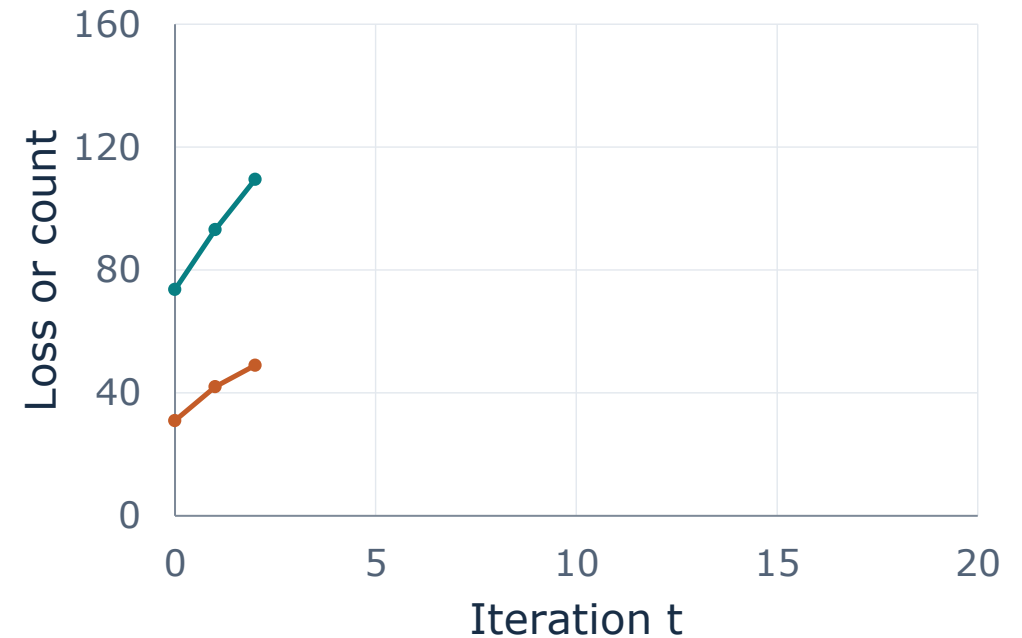
Gradient descent: $\eta = 0.002, t = 2$

Coefficients: $b^{(2)} \approx -5.773416$ $w_1^{(2)} \approx -0.115049$ $w_2^{(2)} \approx 0.966988$



● Class 0 ▲ Class 1 — Decision boundary

$$x_2 = 5.9705 + 0.1190x_1$$

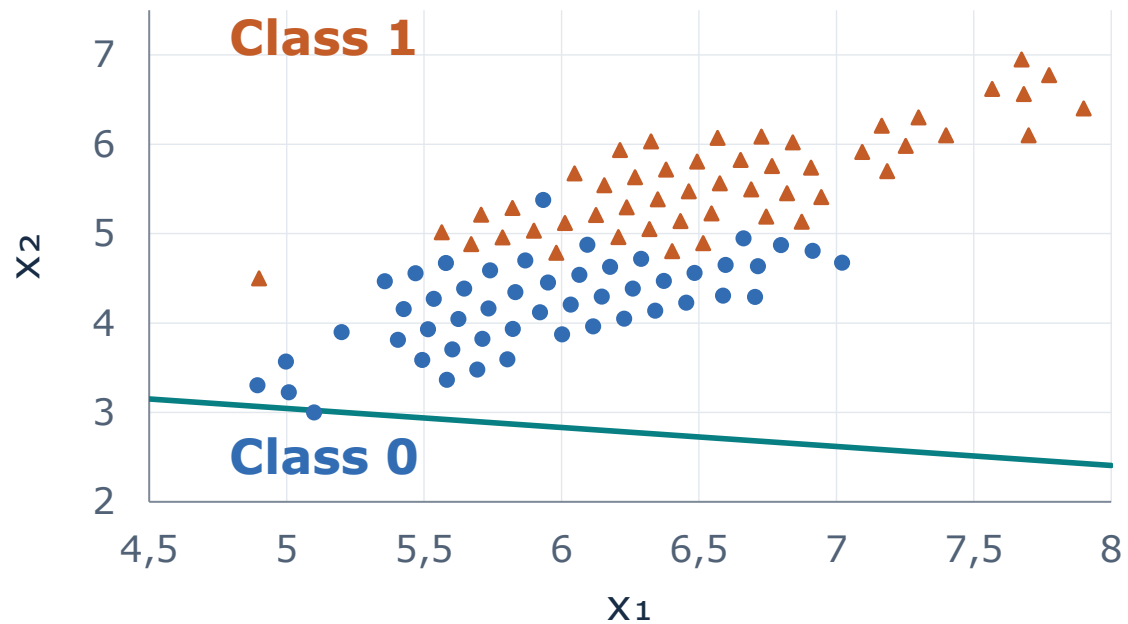


—●— Logistic loss / log 2 —●— Number of mistakes

Logistic loss: 75.928
Mistakes: 49 of 100

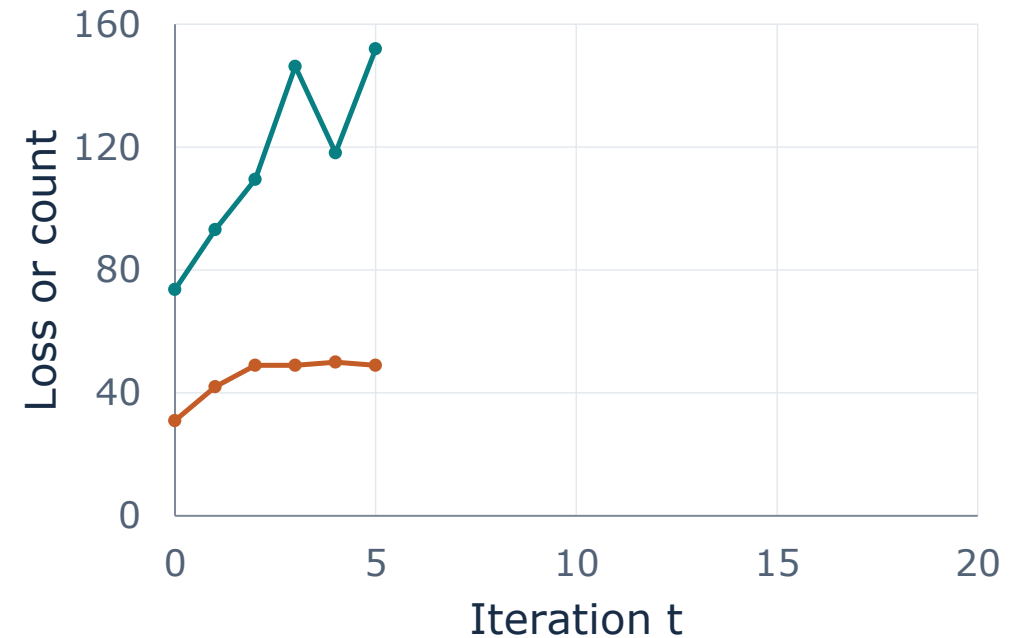
Gradient descent: $\eta = 0.002, t = 5$

Coefficients: $b^{(5)} \approx -5.718821$ $w_1^{(5)} \approx 0.295982$ $w_2^{(5)} \approx 1.392324$



● Class 0 ▲ Class 1 — Decision boundary

$$x_2 = 4.1074 - 0.2126x_1$$

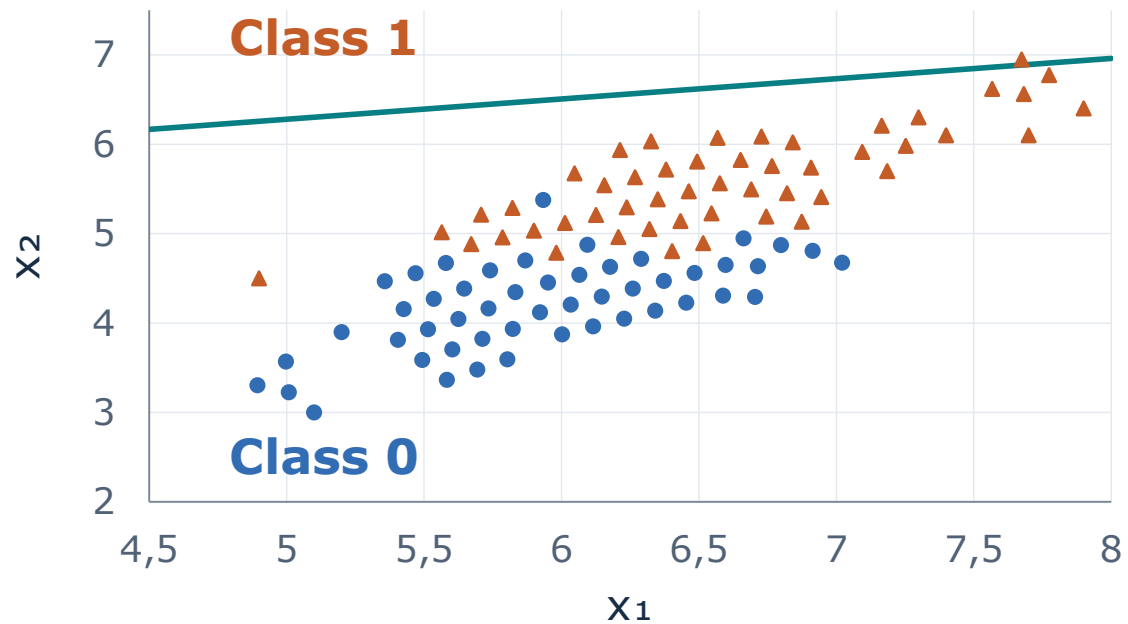


— Logistic loss / log 2 — Number of mistakes

Logistic loss: 105.343
Mistakes: 49 of 100

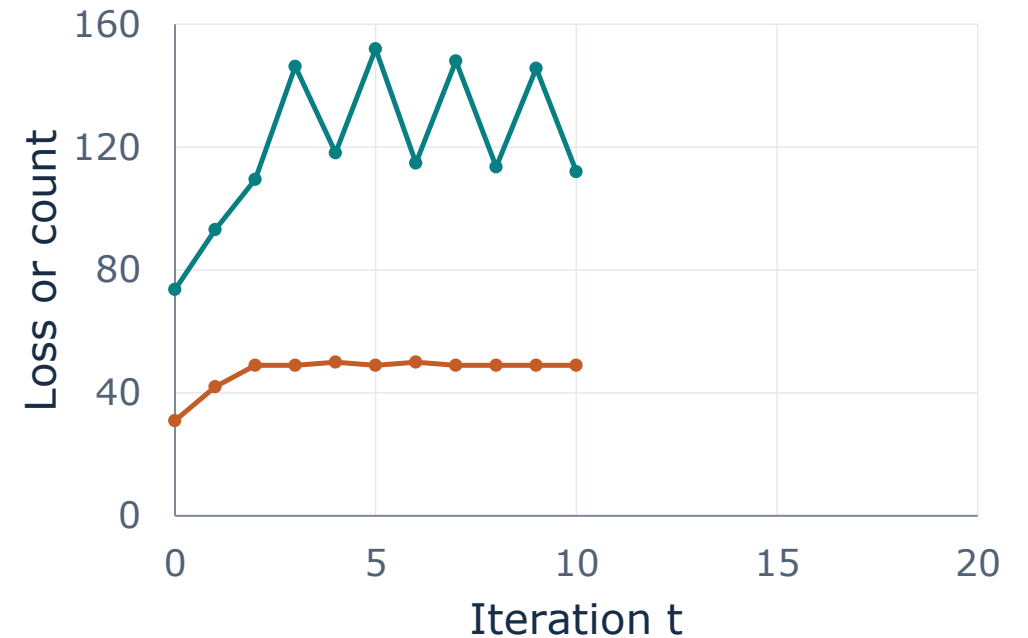
Gradient descent: $\eta = 0.002, t = 10$

Coefficients: $b^{(10)} \approx -5.825631$ $w_1^{(10)} \approx -0.256765$ $w_2^{(10)} \approx 1.132075$



● Class 0 ▲ Class 1 — Decision boundary

$$x_2 = 5.1460 + 0.2268x_1$$

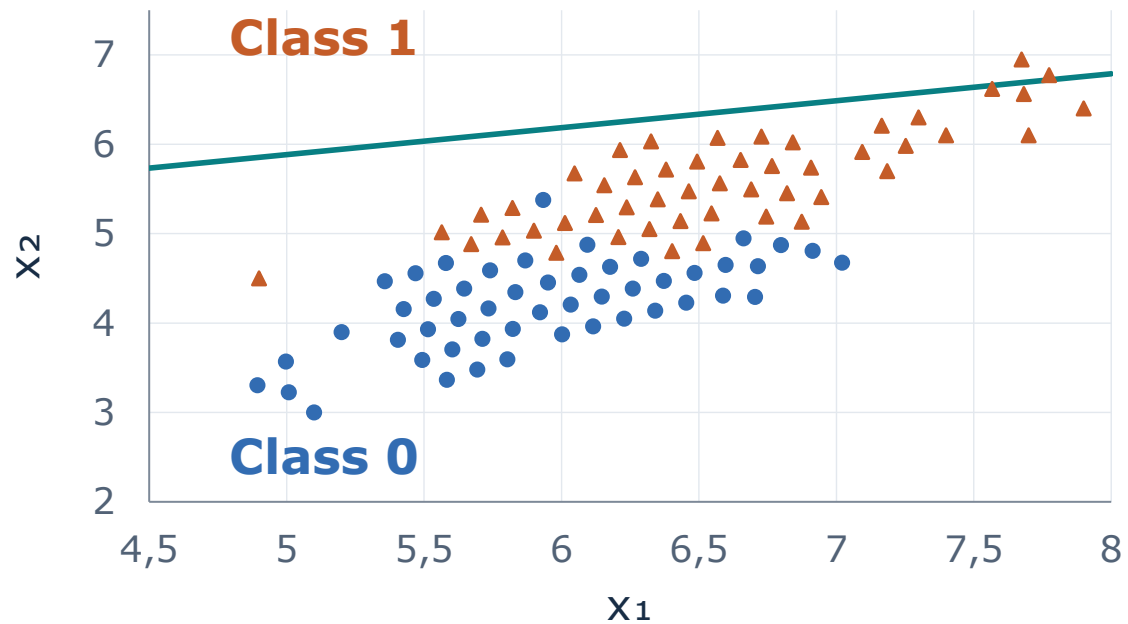


— Logistic loss / log 2 — Number of mistakes

Logistic loss: 77.632
Mistakes: 49 of 100

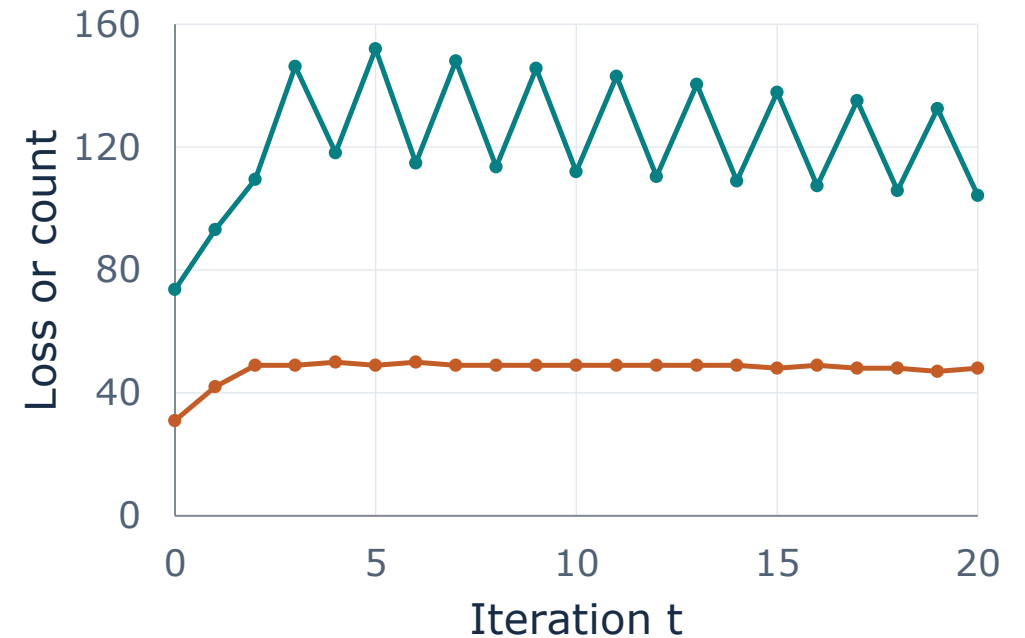
Gradient descent: $\eta = 0.002, t = 20$

Coefficients: $b^{(20)} \approx -5.884803$ $w_1^{(20)} \approx -0.405317$ $w_2^{(20)} \approx 1.344522$



● Class 0 ▲ Class 1 — Decision boundary

$$x_2 = 4.3769 + 0.3015x_1$$

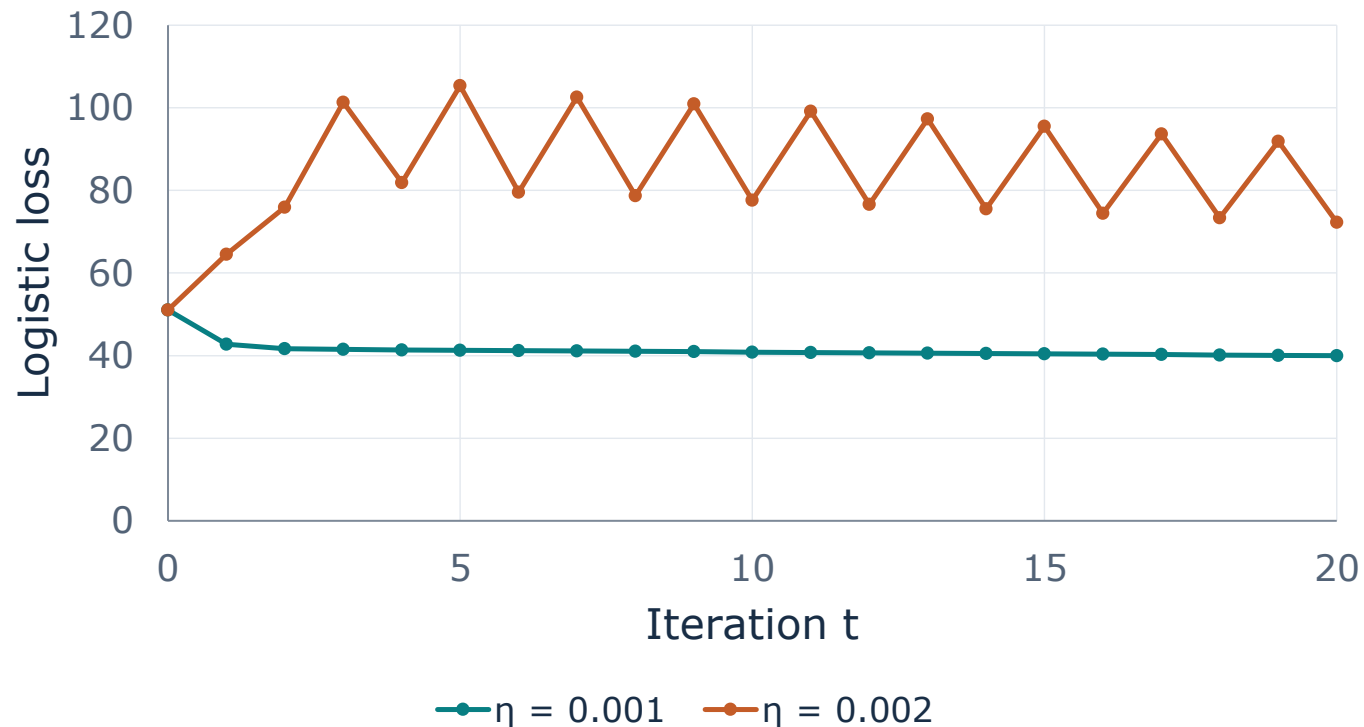


— Logistic loss / log 2 — Number of mistakes

Logistic loss: 72.282
Mistakes: 48 of 100

Step size and logistic loss

Same data and initial coefficients, only the learning rate changes.



Step	Loss $\eta = 0.001$	Loss $\eta = 0.002$
0	51.048	51.048
1	42.733	64.556

**At $\eta = 0.002$,
even the first update
increases the loss.**

A step that is too large can increase the loss.

Today's lecture

Binary classification

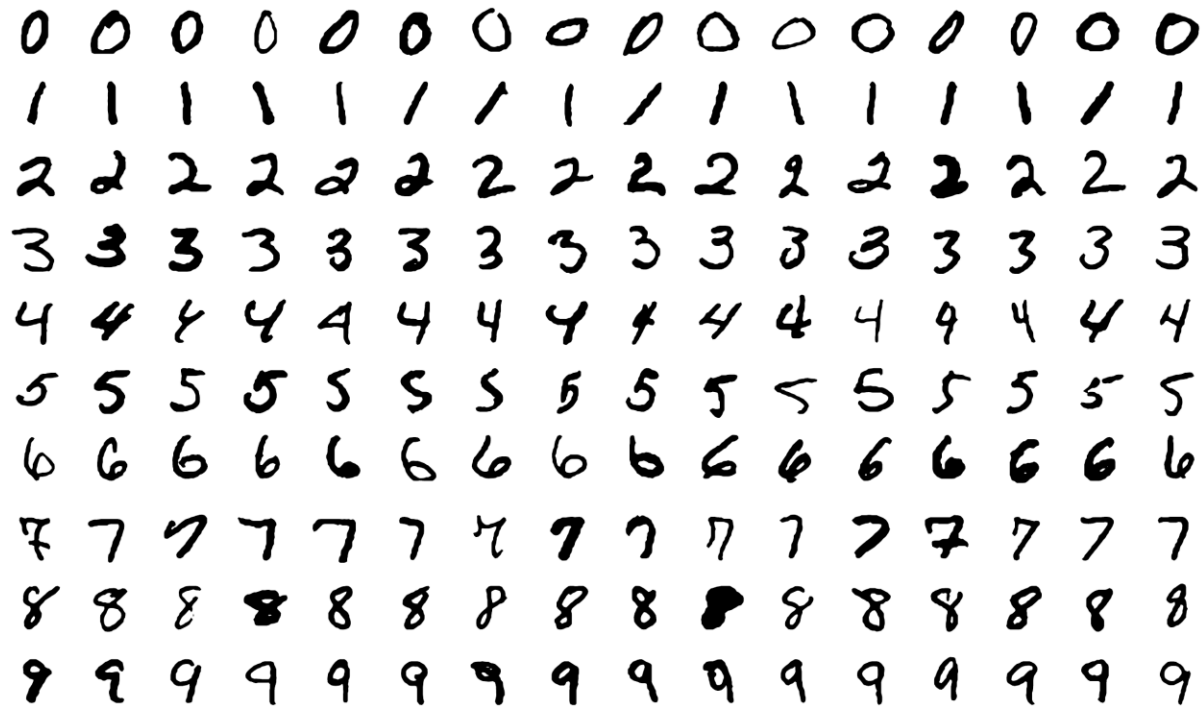
Classification error and logistic loss

Gradient descent

More than two classes

More than two classes

Recognising a handwritten digit means choosing among ten classes: 0, 1, ..., 9.



Each row contains examples of the same digit.

An image is made of pixels.
In greyscale, their intensities
form a matrix of real numbers.

We read the matrix row by row
to form a row vector.

We use the same pixel order
for every image.

$$\mathbf{x} = (x_1, \dots, x_d) \in \mathbb{R}^d$$

Here, d is the number of pixels.

We predict one probability per class

Predicting class probabilities

We number the classes from 1 to K .

For digits 0 to 9, class index k refers to digit $k - 1$.

$$\mathbf{p} : \mathbb{R}^d \rightarrow [0,1]^K$$

$$\mathbf{p}(\mathbf{x}) = (p_1(\mathbf{x}), \dots, p_K(\mathbf{x}))$$

Each component gives the predicted probability of the corresponding class.

For this vector to represent a valid probability distribution,
we require its components to sum to one:

$$\sum_{k=1}^K p_k(\mathbf{x}) = 1$$

If we must choose one label, which should we predict?

The multiclass plug-in rule

Choose the class with the largest predicted probability.

$$h_{\mathbf{p}} : \mathbb{R}^d \rightarrow \{1, \dots, K\}$$

$h_{\mathbf{p}}(\mathbf{x})$ returns the index of the class $k \in \{1, \dots, K\}$
with the largest predicted probability $p_k(\mathbf{x})$.

In case of a tie, we return
the smallest index among those having the largest predicted probability.

How do we build the probability function?

One affine score per class

for each class $k = 1, \dots, K$

$$b_k \in \mathbb{R}, \quad \mathbf{w}_k = \begin{pmatrix} w_{1k} \\ \vdots \\ w_{dk} \end{pmatrix} \in \mathbb{R}^{d \times 1} \quad \boldsymbol{\theta}_k = \begin{pmatrix} b_k \\ \mathbf{w}_k \end{pmatrix} \in \mathbb{R}^{(d+1) \times 1}$$

The k -th score function is defined via

$$f_{\boldsymbol{\theta}_k}(\mathbf{x}) = b_k + \mathbf{x}\mathbf{w}_k$$

As in binary classification, these affine scores are also called **logits**.

Stack all the parameter columns in a **parameter matrix** $\boldsymbol{\Theta}$

$$\boldsymbol{\Theta} = (\boldsymbol{\theta}_1 \quad \dots \quad \boldsymbol{\theta}_K) \in \mathbb{R}^{(d+1) \times K}$$

$$\mathbf{f}_{\boldsymbol{\Theta}}(\mathbf{x}) = \left(f_{\boldsymbol{\theta}_1}(\mathbf{x}), \dots, f_{\boldsymbol{\theta}_K}(\mathbf{x}) \right) \in \mathbb{R}^K$$

Softmax turns scores into probabilities

Warning: separate sigmoids do not ensure a sum of one!!!

Three zero scores would give three probabilities of one half.

Exponentiate each score, then divide by the sum across classes.

$$\text{softmax} : \mathbb{R}^K \rightarrow (0,1)^K$$

$$\mathbf{s} = (s_1, \dots, s_K) \in \mathbb{R}^K$$

$$\text{softmax}(\mathbf{s}) = \left(\frac{e^{s_1}}{\sum_{r=1}^K e^{s_r}}, \dots, \frac{e^{s_K}}{\sum_{r=1}^K e^{s_r}} \right)$$

Positive numbers:

$$[\text{softmax}(\mathbf{s})]_k > 0$$

Valid probabilities:

$$\sum_{k=1}^K [\text{softmax}(\mathbf{s})]_k = \frac{\sum_{k=1}^K e^{s_k}}{\sum_{r=1}^K e^{s_r}} = 1$$

Our multiclass probability model

Apply softmax to the row of affine scores.

$$\mathbf{p}_{\Theta}(\mathbf{x}) = \text{softmax}(\mathbf{f}_{\Theta}(\mathbf{x}))$$

Its component for each class is

$$p_{\Theta,k}(\mathbf{x}) := [\mathbf{p}_{\Theta}(\mathbf{x})]_k = \frac{\exp(f_{\Theta_k}(\mathbf{x}))}{\sum_{r=1}^K \exp(f_{\Theta_r}(\mathbf{x}))}, \quad k = 1, \dots, K$$

Once we have the probability predictor $\mathbf{p}_{\Theta}$

We can build the plug-in classifier $h_{\mathbf{p}_{\Theta}}$

Cross-entropy loss for the observed class

The loss is minus the logarithm of the predicted probability of the observed class

$$\mathbf{p} = (p_1, \dots, p_K), \quad y \in \{1, \dots, K\}, \quad p_y > 0$$

$$\ell_{\text{multi}}(\mathbf{p}, y) = -\log(p_y)$$

Dataset: $(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_N, y_N) \in \mathbb{R}^d \times \{1, \dots, K\}$

Empirical cross-entropy

$$\mathcal{L}_{\text{multi}}(\Theta) = \sum_{i=1}^N \ell_{\text{multi}}(\mathbf{p}_{\Theta}(\mathbf{x}_i), y_i)$$

**Fit the parameter matrix by minimising this summed loss,
e.g., via **gradient descent**.**

$$\text{number of mistakes made by } h_{\mathbf{p}_{\Theta}} \leq \frac{1}{\log(2)} \mathcal{L}_{\text{multi}}(\Theta)$$

Lecture notes and exercises

The notes for today's lecture are available on the course GitHub page.

They explain the material in more detail and include exercises.

I will post the exercise solutions before the next lecture.

github.com/semtem0051/2026_27



Scan for course materials